Real Analysis

Exercise

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Real Analysis: Measure Theory, Integration, and Hilbert Spaces

Chapter 1. Measure Theory

1. Prove that the Cantor set $\mathcal{C}$ constructed in the text is totally disconnected and perfect. In other words, given two distinct points $x, y \in \mathcal{C}$, there is a point $z \notin \mathcal{C}$ that lies between x and y, and yet $\mathcal{C}$ has no isolated points.

[Hint: If $x, y \in \mathcal{C}$ and $|x - y| > 1/3^k$, then x and y belong to two different intervals in C_k. Also, given any $x \in \mathcal{C}$ there is an end-point y_k of some interval in C_k that satisfies $x \neq y_k$ and $|x - y_k| \leq 1/3^k$.]

2. The Cantor set $\mathcal{C}$ can also be described in terms of ternary expansions.

(a) Every number in $[0,1]$ has a ternary expansion

$$x = \sum_{k=1}^{\infty} a_k 3^{-k}, \qquad \text{where } a_k = 0, 1, \text{ or } 2.$$

Note that this decomposition is not unique since, for example, $1/3 = \sum_{k=2}^{\infty} 2/3^k$.

Prove that $x \in \mathcal{C}$ if and only if x has a representation as above where every a_k is either 0 or 2.

(b) The **Cantor-Lebesgue function** is defined on $\mathcal{C}$ by

$$F(x) = \sum_{k=1}^{\infty} \frac{b_k}{2^k} \qquad \text{if } x = \textstyle\sum_{k=1}^{\infty} a_k 3^{-k}, \text{ where } b_k = a_k/2.$$

In this definition, we choose the expansion of x in which $a_k = 0$ or 2.

Show that F is well defined and continuous on $\mathcal{C}$, and moreover $F(0) = 0$ as well as $F(1) = 1$.

(c) Prove that $F : \mathcal{C} \to [0,1]$ is surjective, that is, for every $y \in [0,1]$ there exists $x \in \mathcal{C}$ such that $F(x) = y$.

(d) One can also extend F to be a continuous function on $[0,1]$ as follows. Note that if (a,b) is an open interval of the complement of $\mathcal{C}$, then $F(a) = F(b)$. Hence we may define F to have the constant value $F(a)$ in that interval.

A geometrical construction of F is described in Chapter 3.

3. Cantor sets of constant dissection. Consider the unit interval $[0, 1]$, and let ξ be a fixed real number with $0 < \xi < 1$ (the case $\xi = 1/3$ corresponds to the Cantor set $\mathcal{C}$ in the text).

In stage 1 of the construction, remove the centrally situated open interval in $[0, 1]$ of length ξ. In stage 2, remove two central intervals each of relative length ξ, one in each of the remaining intervals after stage 1, and so on.

Let $\mathcal{C}_\xi$ denote the set which remains after applying the above procedure indefinitely.[6]

(a) Prove that the complement of $\mathcal{C}_\xi$ in $[0, 1]$ is the union of open intervals of total length equal to 1.

(b) Show directly that $m_*(\mathcal{C}_\xi) = 0$.

[Hint: After the kth stage, show that the remaining set has total length $= (1 - \xi)^k$.]

[6]The set we call $\mathcal{C}_\xi$ is sometimes denoted by $\mathcal{C}_{\frac{1-\xi}{2}}$.

4. Cantor-like sets. Construct a closed set $\hat{\mathcal{C}}$ so that at the kth stage of the construction one removes 2^{k-1} centrally situated open intervals each of length ℓ_k, with

$$\ell_1 + 2\ell_2 + \cdots + 2^{k-1}\ell_k < 1.$$

(a) If ℓ_j are chosen small enough, then $\sum_{k=1}^{\infty} 2^{k-1}\ell_k < 1$. In this case, show that $m(\hat{\mathcal{C}}) > 0$, and in fact, $m(\hat{\mathcal{C}}) = 1 - \sum_{k=1}^{\infty} 2^{k-1}\ell_k$.

(b) Show that if $x \in \hat{\mathcal{C}}$, then there exists a sequence of points $\{x_n\}_{n=1}^{\infty}$ such that $x_n \notin \hat{\mathcal{C}}$, yet $x_n \to x$ and $x_n \in I_n$, where I_n is a sub-interval in the complement of $\hat{\mathcal{C}}$ with $|I_n| \to 0$.

(c) Prove as a consequence that $\hat{\mathcal{C}}$ is perfect, and contains no open interval.

(d) Show also that $\hat{\mathcal{C}}$ is uncountable.

5. Suppose E is a given set, and $\mathcal{O}_n$ is the open set:

$$\mathcal{O}_n = \{x : \ d(x, E) < 1/n\}.$$

Show:

(a) If E is compact, then $m(E) = \lim_{n\to\infty} m(\mathcal{O}_n)$.

(b) However, the conclusion in (a) may be false for E closed and unbounded; or E open and bounded.

6. Using translations and dilations, prove the following: Let B be a ball in $\mathbb{R}^d$ of radius r. Then $m(B) = v_d r^d$, where $v_d = m(B_1)$, and B_1 is the unit ball, $B_1 = \{x \in \mathbb{R}^d : \ |x| < 1\}$.

A calculation of the constant v_d is postponed until Exercise 14 in the next chapter.

7. If $\delta = (\delta_1, \ldots, \delta_d)$ is a d-tuple of positive numbers $\delta_i > 0$, and E is a subset of $\mathbb{R}^d$, we define δE by

$$\delta E = \{(\delta_1 x_1, \ldots, \delta_d x_d) : \text{ where } (x_1, \ldots, x_d) \in E\}.$$

Prove that δE is measurable whenever E is measurable, and

$$m(\delta E) = \delta_1 \cdots \delta_d m(E).$$

8. Suppose L is a linear transformation of $\mathbb{R}^d$. Show that if E is a measurable subset of $\mathbb{R}^d$, then so is $L(E)$, by proceeding as follows:

(a) Note that if E is compact, so is $L(E)$. Hence if E is an F_σ set, so is $L(E)$.

(b) Because L automatically satisfies the inequality

$$|L(x) - L(x')| \leq M|x - x'|$$

for some M, we can see that L maps any cube of side length ℓ into a cube of side length $c_d M\ell$, with $c_d = 2\sqrt{d}$. Now if $m(E) = 0$, there is a collection of cubes $\{Q_j\}$ such that $E \subset \bigcup_j Q_j$, and $\sum_j m(Q_j) < \epsilon$. Thus $m_*(L(E)) \leq c'\epsilon$, and hence $m(L(E)) = 0$. Finally, use Corollary 3.5.

One can show that $m(L(E)) = |\det L|\, m(E)$; see Problem 4 in the next chapter.

9. Give an example of an open set $\mathcal{O}$ with the following property: the boundary of the closure of $\mathcal{O}$ has positive Lebesgue measure.

[Hint: Consider the set obtained by taking the union of open intervals which are deleted at the odd steps in the construction of a Cantor-like set.]

10. This exercise provides a construction of a decreasing sequence of positive continuous functions on the interval $[0, 1]$, whose pointwise limit is *not* Riemann integrable.

Let $\hat{\mathcal{C}}$ denote a Cantor-like set obtained from the construction detailed in Exercise 4, so that in particular $m(\hat{\mathcal{C}}) > 0$. Let F_1 denote a piecewise-linear and continuous function on $[0, 1]$, with $F_1 = 1$ in the complement of the first interval removed in the construction of $\hat{\mathcal{C}}$, $F_1 = 0$ at the center of this interval, and $0 \leq F_1(x) \leq 1$ for all x. Similarly, construct $F_2 = 1$ in the complement of the intervals in stage two of the construction of $\hat{\mathcal{C}}$, with $F_2 = 0$ at the center of these intervals, and $0 \leq F_2 \leq 1$. Continuing this way, let $f_n = F_1 \cdot F_2 \cdots F_n$ (see Figure 5).

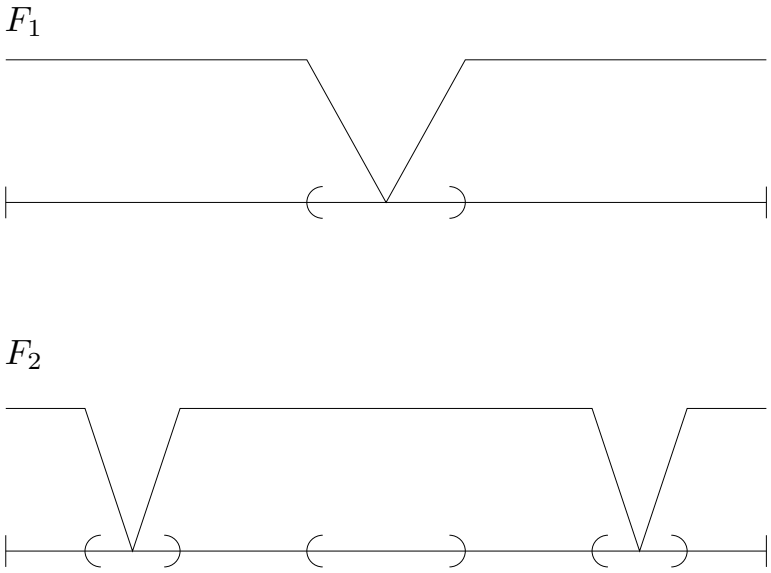


Figure 5. Construction of $\{F_n\}$ in Exercise 10

Prove the following:

(a) For all $n \geq 1$ and all $x \in [0, 1]$, one has $0 \leq f_n(x) \leq 1$ and $f_n(x) \geq f_{n+1}(x)$. Therefore, $f_n(x)$ converges to a limit as $n \to \infty$ which we denote by $f(x)$.

(b) The function f is discontinuous at every point of $\hat{\mathcal{C}}$.

[Hint: Note that $f(x) = 1$ if $x \in \hat{\mathcal{C}}$, and find a sequence of points $\{x_n\}$ so that $x_n \to x$ and $f(x_n) = 0$.]

Now $\int f_n(x)\,dx$ is decreasing, hence $\int f_n$ converges. However, a bounded function is Riemann integrable if and only if its set of discontinuities has measure zero.

(The proof of this fact, which is given in the Appendix of Book I, is outlined in Problem 4.) Since f is discontinuous on a set of positive measure, we find that f is not Riemann integrable.

11. Let A be the subset of $[0, 1]$ which consists of all numbers which do not have the digit 4 appearing in their decimal expansion. Find $m(A)$.

12. Theorem 1.3 states that every open set in $\mathbb{R}$ is the disjoint union of open intervals. The analogue in $\mathbb{R}^d$, $d \geq 2$, is generally false. Prove the following:

(a) An open disc in $\mathbb{R}^2$ is not the disjoint union of open rectangles.

[Hint: What happens to the boundary of any of these rectangles?]

(b) An open connected set Ω is the disjoint union of open rectangles if and only if Ω is itself an open rectangle.

13. The following deals with G_δ and F_σ sets.

(a) Show that a closed set is a G_δ and an open set an F_σ.
[Hint: If F is closed, consider $\mathcal{O}_n = \{x : d(x, F) < 1/n\}$.]

(b) Give an example of an F_σ which is not a G_δ.
[Hint: This is more difficult; let F be a denumerable set that is dense.]

(c) Give an example of a Borel set which is not a G_δ nor an F_σ.

14. The purpose of this exercise is to show that covering by a *finite* number of intervals will not suffice in the definition of the outer measure m_*.

The **outer Jordan content** $J_*(E)$ of a set E in $\mathbb{R}$ is defined by

$$J_*(E) = \inf \sum_{j=1}^{N} |I_j|,$$

where the inf is taken over every *finite* covering $E \subset \bigcup_{j=1}^{N} I_j$, by intervals I_j.

(a) Prove that $J_*(E) = J_*(\overline{E})$ for every set E (here $\overline{E}$ denotes the closure of E).

(b) Exhibit a countable subset $E \subset [0,1]$ such that $J_*(E) = 1$ while $m_*(E) = 0$.

15. At the start of the theory, one might define the outer measure by taking coverings by rectangles instead of cubes. More precisely, we define

$$m_*^{\mathcal{R}}(E) = \inf \sum_{j=1}^{\infty} |R_j|,$$

where the inf is now taken over all countable coverings $E \subset \bigcup_{j=1}^{\infty} R_j$ by (closed) rectangles.

Show that this approach gives rise to the same theory of measure developed in the text, by proving that $m_*(E) = m_*^{\mathcal{R}}(E)$ for every subset E of $\mathbb{R}^d$.

[Hint: Use Lemma 1.1.]

16. The Borel-Cantelli lemma. Suppose $\{E_k\}_{k=1}^{\infty}$ is a countable family of measuable subsets of $\mathbb{R}^d$ and that

$$\sum_{k=1}^{\infty} m(E_k) < \infty.$$

Let

$$\begin{aligned} E &= \{x \in \mathbb{R}^d : x \in E_k, \text{ for infinitely many } k\} \\ &= \limsup_{k\to\infty}(E_k). \end{aligned}$$

(a) Show that E is measurable.

(b) Prove $m(E) = 0$.

[Hint: Write $E = \bigcap_{n=1}^{\infty} \bigcup_{k\geq n} E_k$.]

17. Let $\{f_n\}$ be a sequence of measurable functions on $[0,1]$ with $|f_n(x)| < \infty$ for a.e x. Show that there exists a sequence c_n of positive real numbers such that

$$\frac{f_n(x)}{c_n} \to 0 \qquad \text{a.e. } x$$

[Hint: Pick c_n such that $m(\{x : |f_n(x)/c_n| > 1/n\}) < 2^{-n}$, and apply the Borel-Cantelli lemma.]

18. Prove the following assertion: Every measurable function is the limit a.e. of a sequence of continuous functions.

19. Here are some observations regarding the set operation $A + B$.

(a) Show that if either A and B is open, then $A + B$ is open.

(b) Show that if A and B are closed, then $A + B$ is measurable.

(c) Show, however, that $A + B$ might not be closed even though A and B are closed.

[Hint: For (b) show that $A + B$ is an F_σ set.]

20. Show that there exist closed sets A and B with $m(A) = m(B) = 0$, but $m(A + B) > 0$:

(a) In $\mathbb{R}$, let $A = \mathcal{C}$ (the Cantor set), $B = \mathcal{C}/2$. Note that $A + B \supset [0, 1]$.

(b) In $\mathbb{R}^2$, observe that if $A = I \times \{0\}$ and $B = \{0\} \times I$ (where $I = [0, 1]$), then $A + B = I \times I$.

21. Prove that there is a continuous function that maps a Lebesgue measurable set to a non-measurable set.

[Hint: Consider a non-measurable subset of $[0, 1]$, and its inverse image in $\mathcal{C}$ by the function F in Exercise 2.]

22. Let $\chi_{[0,1]}$ be the characteristic function of $[0,1]$. Show that there is no everywhere continuous function f on $\mathbb{R}$ such that

$$f(x) = \chi_{[0,1]}(x) \qquad \text{almost everywhere.}$$

23. Suppose $f(x, y)$ is a function on $\mathbb{R}^2$ that is separately continuous: for each fixed variable, f is continuous in the other variable. Prove that f is measurable on $\mathbb{R}^2$.

[Hint: Approximate f in the variable x by piecewise-linear functions f_n so that $f_n \to f$ pointwise.]

24. Does there exist an enumeration $\{r_n\}_{n=1}^{\infty}$ of the rationals, such that the complement of

$$\bigcup_{n=1}^{\infty}\left(r_n - \frac{1}{n}, r_n + \frac{1}{n}\right)$$

in $\mathbb{R}$ is non-empty?

[Hint: Find an enumeration where the only rationals outside of a fixed bounded interval take the form r_n, with $n = m^2$ for some integer m.]

25. An alternative definition of measurability is as follows: E is measurable if for every $\epsilon > 0$ there is a *closed* set F contained in E with $m_*(E - F) < \epsilon$. Show that this definition is equivalent with the one given in the text.

26. Suppose $A \subset E \subset B$, where A and B are measurable sets of finite measure. Prove that if $m(A) = m(B)$, then E is measurable.

27. Suppose E_1 and E_2 are a pair of compact sets in $\mathbb{R}^d$ with $E_1 \subset E_2$, and let $a = m(E_1)$ and $b = m(E_2)$. Prove that for any c with $a < c < b$, there is a compact set E with $E_1 \subset E \subset E_2$ and $m(E) = c$.

[Hint: As an example, if $d = 1$ and E is a measurable subset of $[0, 1]$, consider $m(E \cap [0, t])$ as a function of t.]

28. Let E be a subset of $\mathbb{R}$ with $m_*(E) > 0$. Prove that for each $0 < \alpha < 1$, there exists an open interval I so that

$$m_*(E \cap I) \geq \alpha\, m_*(I).$$

Loosely speaking, this estimate shows that E contains almost a whole interval.

[Hint: Choose an open set $\mathcal{O}$ that contains E, and such that $m_*(E) \geq \alpha\, m_*(\mathcal{O})$. Write $\mathcal{O}$ as the countable union of disjoint open intervals, and show that one of these intervals must satisfy the desired property.]

29. Suppose E is a measurable subset of $\mathbb{R}$ with $m(E) > 0$. Prove that the **difference set** of E, which is defined by

$$\{z \in \mathbb{R} : \ z = x - y \text{ for some } x, y \in E\},$$

contains an open interval centered at the origin.

If E contains an interval, then the conclusion is straightforward. In general, one may rely on Exercise 28.

[Hint: Indeed, by Exercise 28, there exists an open interval I so that $m(E \cap I) \geq (9/10)\, m(I)$. If we denote $E \cap I$ by E_0, and suppose that the difference set of E_0 does not contain an open interval around the origin, then for arbitrarily small a the sets E_0, and $E_0 + a$ are disjoint. From the fact that $(E_0 \cup (E_0 + a)) \subset (I \cup (I + a))$ we get a contradiction, since the left-hand side has measure $2m(E_0)$, while the right-hand side has measure only slightly larger than $m(I)$.]

A more general formulation of this result is as follows.

30. If E and F are measurable, and $m(E) > 0$, $m(F) > 0$, prove that

$$E + F = \{x + y : x \in E,\ x \in F\}$$

contains an interval.

31. The result in Exercise 29 provides an alternate proof of the non-measurability of the set $\mathcal{N}$ studied in the text. In fact, we may also prove the non-measurability of a set in $\mathbb{R}$ that is very closely related to the set $\mathcal{N}$.

Given two real numbers x and y, we shall write as before that $x \sim y$ whenever the difference $x - y$ is rational. Let $\mathcal{N}^*$ denote a set that consists of one element in each equivalence class of $\sim$. Prove that $\mathcal{N}^*$ is non-measurable by using the result in Exercise 29.

[Hint: If $\mathcal{N}^*$ is measurable, then so are its translates $\mathcal{N}_n^* = \mathcal{N}^* + r_n$, where $\{r_n\}_{n=1}^{\infty}$ is an enumeration of $\mathbb{Q}$. How does this imply that $m(\mathcal{N}^*) > 0$? Can the difference set of $\mathcal{N}^*$ contain an open interval centered at the origin?]

32. Let $\mathcal{N}$ denote the non-measurable subset of $I = [0, 1]$ constructed at the end of Section 3.

(a) Prove that if E is a measurable subset of $\mathcal{N}$, then $m(E) = 0$.

(b) If G is a subset of $\mathbb{R}$ with $m_*(G) > 0$, prove that a subset of G is non-measurable.

[Hint: For (a) use the translates of E by the rationals.]

33. Let $\mathcal{N}$ denote the non-measurable set constructed in the text. Recall from the exercise above that measurable subsets of $\mathcal{N}$ have measure zero.

Show that the set $\mathcal{N}^c = I - \mathcal{N}$ satisfies $m_*(\mathcal{N}^c) = 1$, and conclude that if $E_1 = \mathcal{N}$ and $E_2 = \mathcal{N}^c$, then

$$m_*(E_1) + m_*(E_2) \neq m_*(E_1 \cup E_2),$$

although E_1 and E_2 are disjoint.

[Hint: To prove that $m_*(\mathcal{N}^c) = 1$, argue by contradiction and pick a measurable set U such that $U \subset I$, $\mathcal{N}^c \subset U$ and $m_*(U) < 1 - \epsilon$.]

34. Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be any two Cantor sets (constructed in Exercise 3). Show that there exists a function $F : [0,1] \to [0,1]$ with the following properties:

(i) F is continuous and bijective,

(ii) F is monotonically increasing,

(iii) F maps $\mathcal{C}_1$ surjectively onto $\mathcal{C}_2$.

[Hint: Copy the construction of the standard Cantor-Lebesgue function.]

35. Give an example of a measurable function f and a continuous function Φ so that $f \circ \Phi$ is non-measurable.

[Hint: Let $\Phi : \mathcal{C}_1 \to \mathcal{C}_2$ as in Exercise 34, with $m(\mathcal{C}_1) > 0$ and $m(\mathcal{C}_2) = 0$. Let $N \subset \mathcal{C}_1$ be non-measurable, and take $f = \chi_{\Phi(N)}$.]

Use the construction in the hint to show that there exists a Lebesgue measurable set that is not a Borel set.

36. This exercise provides an example of a measurable function f on $[0,1]$ such that every function g equivalent to f (in the sense that f and g differ only on a set of measure zero) is discontinuous at *every* point.

(a) Construct a measurable set $E \subset [0,1]$ such that for any non-empty open sub-interval I in $[0,1]$, both sets $E \cap I$ and $E^c \cap I$ have positive measure.

(b) Show that $f = \chi_E$ has the property that whenever $g(x) = f(x)$ a.e x, then g must be discontinuous at every point in $[0,1]$.

[Hint: For the first part, consider a Cantor-like set of positive measure, and add in each of the intervals that are omitted in the first step of its construction, another Cantor-like set. Continue this procedure indefinitely.]

37. Suppose Γ is a curve $y = f(x)$ in $\mathbb{R}^2$, where f is continuous. Show that $m(\Gamma) = 0$.

[Hint: Cover Γ by rectangles, using the uniform continuity of f.]

38. Prove that $(a+b)^\gamma \geq a^\gamma + b^\gamma$ whenever $\gamma \geq 1$ and $a, b \geq 0$. Also, show that the reverse inequality holds when $0 \leq \gamma \leq 1$.

[Hint: Integrate the inequality between $(a+t)^{\gamma-1}$ and $t^{\gamma-1}$ from 0 to b.]

39. Establish the inequality

$$\frac{x_1 + \cdots + x_d}{d} \geq (x_1 \cdots x_d)^{1/d} \qquad \text{for all } x_j \geq 0,\ j = 1, \ldots, d \tag{10}$$

by using backward induction as follows:

(a) The inequality is true whenever d is a power of 2 ($d = 2^k$, $k \geq 1$).

(b) If (10) holds for some integer $d \geq 2$, then it must hold for $d - 1$, that is, one has $(y_1 + \cdots + y_{d-1})/(d-1) \geq (y_1 \cdots y_{d-1})^{1/(d-1)}$ for all $y_j \geq 0$, with $j = 1, \ldots, d-1$.

[Hint: For (a), if $k \geq 2$, write $(x_1 + \cdots + x_{2^k})/2^k$ as $(A + B)/2$, where $A = (x_1 + \cdots + x_{2^{k-1}})/2^{k-1}$, and apply the inequality when $d = 2$. For (b), apply the inequality to $x_1 = y_1, \ldots, x_{d-1} = y_{d-1}$ and $x_d = (y_1 + \cdots + y_{d-1})/(d-1)$.]

Chapter 2. Integration Theory

1. Given a collection of sets $F_1, F_2, \ldots, F_n$, construct another collection $F_1^*, F_2^*, \ldots, F_N^*$, with $N = 2^n - 1$, so that $\bigcup_{k=1}^n F_k = \bigcup_{j=1}^N F_j^*$; the collection $\{F_j^*\}$ is disjoint; also

$F_k = \bigcup_{F_j^* \subset F_k} F_j^*$, for every k.

[Hint: Consider the 2^n sets $F_1' \cap F_2' \cap \cdots \cap F_n'$ where each F_k' is either F_k or F_k^c.]

2. In analogy to Proposition 2.5, prove that if f is integrable on $\mathbb{R}^d$ and $\delta > 0$, then $f(\delta x)$ converges to $f(x)$ in the L^1-norm as $\delta \to 1$.

3. Suppose f is integrable on $(-\pi, \pi]$ and extended to $\mathbb{R}$ by making it periodic of period 2π. Show that

$$\int_{-\pi}^{\pi} f(x)\,dx = \int_{I} f(x)\,dx,$$

where I is any interval in $\mathbb{R}$ of length 2π.

[Hint: I is contained in two consecutive intervals of the form $(k\pi, (k+2)\pi)$.]

4. Suppose f is integrable on $[0, b]$, and

$$g(x) = \int_x^b \frac{f(t)}{t}\, dt \qquad \text{for } 0 < x \leq b.$$

Prove that g is integrable on $[0, b]$ and

$$\int_0^b g(x)\, dx = \int_0^b f(t)\, dt.$$

5. Suppose F is a closed set in $\mathbb{R}$, whose complement has finite measure, and let $\delta(x)$ denote the distance from x to F, that is,

$$\delta(x) = d(x, F) = \inf\{|x - y| : \ y \in F\}.$$

Consider

$$I(x) = \int_{\mathbb{R}} \frac{\delta(y)}{|x - y|^2}\, dy.$$

(a) Prove that δ is continuous, by showing that it satisfies the Lipschitz condition

$$|\delta(x) - \delta(y)| \leq |x - y|.$$

(b) Show that $I(x) = \infty$ for each $x \notin F$.

(c) Show that $I(x) < \infty$ for a.e. $x \in F$. This may be surprising in view of the fact that the Lispshitz condition cancels only one power of $|x - y|$ in the integrand of I.

[Hint: For the last part, investigate $\int_F I(x)\, dx$.]

6. Integrability of f on $\mathbb{R}$ does not necessarily imply the convergence of $f(x)$ to 0 as $x \to \infty$.

(a) There exists a positive continuous function f on $\mathbb{R}$ so that f is integrable on $\mathbb{R}$, but yet $\limsup_{x\to\infty} f(x) = \infty$.

(b) However, if we assume that f is uniformly continuous on $\mathbb{R}$ and integrable, then $\lim_{|x|\to\infty} f(x) = 0$.

[Hint: For (a), construct a continuous version of the function equal to n on the segment $[n, n + 1/n^3)$, $n \geq 1$.]

7. Let $\Gamma \subset \mathbb{R}^d \times \mathbb{R}$, $\Gamma = \{(x, y) \in \mathbb{R}^d \times \mathbb{R} : \ y = f(x)\}$, and assume f is measurable on $\mathbb{R}^d$. Show that Γ is a measurable subset of $\mathbb{R}^{d+1}$, and $m(\Gamma) = 0$.

8. If f is integrable on $\mathbb{R}$, show that $F(x) = \int_{-\infty}^{x} f(t)\,dt$ is uniformly continuous.

9. Tchebychev inequality. Suppose $f \geq 0$, and f is integrable. If $\alpha > 0$ and $E_\alpha = \{x : f(x) > \alpha\}$, prove that

$$m(E_\alpha) \leq \frac{1}{\alpha} \int f.$$

10. Suppose $f \geq 0$, and let $E_{2^k} = \{x : f(x) > 2^k\}$ and $F_k = \{x : 2^k < f(x) \leq 2^{k+1}\}$. If f is finite almost everywhere, then

$$\bigcup_{k=-\infty}^{\infty} F_k = \{f(x) > 0\},$$

and the sets F_k are disjoint.

Prove that f is integrable if and only if

$$\sum_{k=-\infty}^{\infty} 2^k m(F_k) < \infty, \qquad \text{if and only if} \qquad \sum_{k=-\infty}^{\infty} 2^k m(E_{2^k}) < \infty.$$

Use this result to verify the following assertions. Let

$$f(x) = \begin{cases} |x|^{-a} & \text{if } |x| \leq 1, \\ 0 & \text{otherwise,} \end{cases} \qquad \text{and} \qquad g(x) = \begin{cases} |x|^{-b} & \text{if } |x| > 1, \\ 0 & \text{otherwise.} \end{cases}$$

Then f is integrable on $\mathbb{R}^d$ if and only if $a < d$; also g is integrable on $\mathbb{R}^d$ if and only if $b > d$.

11. Prove that if f is integrable on $\mathbb{R}^d$, real-valued, and $\int_E f(x)\,dx \geq 0$ for every measurable E, then $f(x) \geq 0$ a.e. x. As a result, if $\int_E f(x)\,dx = 0$ for every measurable E, then $f(x) = 0$ a.e.

12. Show that there are $f \in L^1(\mathbb{R}^d)$ and a sequence $\{f_n\}$ with $f_n \in L^1(\mathbb{R}^d)$ such that

$$\|f - f_n\|_{L^1} \to 0,$$

but $f_n(x) \to f(x)$ for no x.

[Hint: In $\mathbb{R}$, let $f_n = \chi_{I_n}$, where I_n is an appropriately chosen sequence of intervals with $m(I_n) \to 0$.]

13. Give an example of two measurable sets A and B such that $A + B$ is not measurable.

[Hint: In $\mathbb{R}^2$ take $A = \{0\} \times [0, 1]$ and $B = \mathcal{N} \times \{0\}$.]

14. In Exercise 6 of the previous chapter we saw that $m(B) = v_d r^d$, whenever B is a ball of radius r in $\mathbb{R}^d$ and $v_d = m(B_1)$, with B_1 the unit ball. Here we evaluate the constant v_d.

(a) For $d = 2$, prove using Corollary 3.8 that

$$v_2 = 2 \int_{-1}^{1} (1 - x^2)^{1/2} \, dx,$$

and hence by elementary calculus, that $v_2 = \pi$.

(b) By similar methods, show that

$$v_d = 2v_{d-1} \int_{0}^{1} (1 - x^2)^{(d-1)/2} \, dx.$$

(c) The result is

$$v_d = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)}.$$

Another derivation is in Exercise 5 in Chapter 6 below. Relevant facts about the gamma and beta functions can be found in Chapter 6 of Book II.

15. Consider the function defined over $\mathbb{R}$ by

$$f(x) = \begin{cases} x^{-1/2} & \text{if } 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

For a fixed enumeration $\{r_n\}_{n=1}^{\infty}$ of the rationals $\mathbb{Q}$, let

$$F(x) = \sum_{n=1}^{\infty} 2^{-n} f(x - r_n).$$

Prove that F is integrable, hence the series defining F converges for almost every $x \in \mathbb{R}$. However, observe that this series is unbounded on every interval, and in fact, any function $\tilde{F}$ that agrees with F a.e is unbounded in any interval.

16. Suppose f is integrable on $\mathbb{R}^d$. If $\delta = (\delta_1, \ldots, \delta_d)$ is a d-tuple of non-zero real numbers, and

$$f^\delta(x) = f(\delta x) = f(\delta_1 x_1, \ldots, \delta_d x_d),$$

show that f^δ is integrable with

$$\int_{\mathbb{R}^d} f^\delta(x)\, dx = |\delta_1|^{-1} \cdots |\delta_d|^{-1} \int_{\mathbb{R}^d} f(x)\, dx.$$

17. Suppose f is defined on $\mathbb{R}^2$ as follows: $f(x,y) = a_n$ if $n \le x < n+1$ and $n \le y < n+1$, $(n \ge 0)$; $f(x,y) = -a_n$ if $n \le x < n+1$ and $n+1 \le y < n+2$, $(n \ge 0)$; while $f(x,y) = 0$ elsewhere. Here $a_n = \sum_{k \le n} b_k$, with $\{b_k\}$ a positive sequence such that $\sum_{k=0}^{\infty} b_k = s < \infty$.

(a) Verify that each slice f^y and f_x is integrable. Also for all x, $\int f_x(y)\,dy = 0$, and hence $\int \left(\int f(x,y)\,dy \right)\,dx = 0$.

(b) However, $\int f^y(x)\,dx = a_0$ if $0 \le y < 1$, and $\int f^y(x)\,dx = a_n - a_{n-1}$ if $n \le y < n+1$ with $n \ge 1$. Hence $y \mapsto \int f^y(x)\,dx$ is integrable on $(0,\infty)$ and

$$\int \left(\int f(x,y)\,dx \right)\,dy = s.$$

(c) Note that $\int_{\mathbb{R}\times\mathbb{R}} |f(x,y)|\,dx\,dy = \infty$.

18. Let f be a measurable finite-valued function on $[0,1]$, and suppose that $|f(x) - f(y)|$ is integrable on $[0,1] \times [0,1]$. Show that $f(x)$ is integrable on $[0,1]$.

19. Suppose f is integrable on $\mathbb{R}^d$. For each $\alpha > 0$, let $E_\alpha = \{x : |f(x)| > \alpha\}$. Prove that

$$\int_{\mathbb{R}^d} |f(x)|\, dx = \int_0^\infty m(E_\alpha)\, d\alpha.$$

20. The problem (highlighted in the discussion preceding Fubini's theorem) that certain slices of measurable sets can be non-measurable may be avoided by restricting attention to Borel measurable functions and Borel sets. In fact, prove the following:

Suppose E is a Borel set in $\mathbb{R}^2$. Then for every y, the slice E^y is a Borel set in $\mathbb{R}$.

[Hint: Consider the collection $\mathcal{C}$ of subsets E of $\mathbb{R}^2$ with the property that each slice E^y is a Borel set in $\mathbb{R}$. Verify that $\mathcal{C}$ is a σ-algebra that contains the open sets.]

21. Suppose that f and g are measurable functions on $\mathbb{R}^d$.

(a) Prove that $f(x-y)g(y)$ is measurable on $\mathbb{R}^{2d}$.

(b) Show that if f and g are integrable on $\mathbb{R}^d$, then $f(x-y)g(y)$ is integrable on $\mathbb{R}^{2d}$.

(c) Recall the definition of the convolution of f and g given by

$$(f*g)(x) = \int_{\mathbb{R}^d} f(x-y)g(y)\,dy.$$

Show that $f*g$ is well defined for a.e. x (that is, $f(x-y)g(y)$ is integrable on $\mathbb{R}^d$ for a.e. x).

(d) Show that $f*g$ is integrable whenever f and g are integrable, and that

$$\|f*g\|_{L^1(\mathbb{R}^d)} \leq \|f\|_{L^1(\mathbb{R}^d)}\,\|g\|_{L^1(\mathbb{R}^d)},$$

with equality if f and g are non-negative.

(e) The Fourier transform of an integrable function f is defined by

$$\hat{f}(\xi) = \int_{\mathbb{R}^d} f(x)e^{-2\pi i x\cdot\xi}\,dx.$$

Check that $\hat{f}$ is bounded and is a continuous function of ξ. Prove that for each ξ one has

$$\widehat{(f*g)}(\xi) = \hat{f}(\xi)\hat{g}(\xi).$$

22. Prove that if $f \in L^1(\mathbb{R}^d)$ and

$$\hat{f}(\xi) = \int_{\mathbb{R}^d} f(x) e^{-2\pi i x\xi}\, dx,$$

then $\hat{f}(\xi) \to 0$ as $|\xi| \to \infty$. (This is the Riemann-Lebesgue lemma.)
[Hint: Write $\hat{f}(\xi) = \frac{1}{2}\int_{\mathbb{R}^d}[f(x) - f(x-\xi')]e^{-2\pi i x\xi}\, dx$, where $\xi' = \frac{1}{2}\frac{\xi}{|\xi|^2}$, and use Proposition 2.5.]

23. As an application of the Fourier transform, show that there does not exist a function $I \in L^1(\mathbb{R}^d)$ such that

$$f * I = f \qquad \text{for all } f \in L^1(\mathbb{R}^d).$$

24. Consider the convolution

$$(f * g)(x) = \int_{\mathbb{R}^d} f(x-y)g(y)\,dy.$$

(a) Show that $f * g$ is uniformly continuous when f is integrable and g bounded.

(b) If in addition g is integrable, prove that $(f * g)(x) \to 0$ as $|x| \to \infty$.

25. Show that for each $\epsilon > 0$ the function $F(\xi) = \frac{1}{(1+|\xi|^2)^\epsilon}$ is the Fourier transform of an L^1 function.
[Hint: With $K_\delta(x) = e^{-\pi|x|^2/\delta}\delta^{-d/2}$ consider $f(x) = \int_0^\infty K_\delta(x)e^{-\pi\delta}\delta^{\epsilon-1}\,d\delta$. Use Fubini's theorem to prove $f \in L^1(\mathbb{R}^d)$, and

$$\hat{f}(\xi) = \int_0^\infty e^{-\pi\delta|\xi|^2}e^{-\pi\delta}\delta^{\epsilon-1}\,d\delta,$$

and evaluate the last integral as $\pi^{-\epsilon}\Gamma(\epsilon)\frac{1}{(1+|\xi|^2)^\epsilon}$. Here $\Gamma(s)$ is the gamma function defined by $\Gamma(s) = \int_0^\infty e^{-t}t^{s-1}\,dt$.]

Chapter 3. Differentiation and Integration

1. Suppose φ is an integrable function on $\mathbb{R}^d$ with $\int_{\mathbb{R}^d} \varphi(x)\,dx = 1$. Set $K_\delta(x) = \delta^{-d}\varphi(x/\delta)$, $\delta > 0$.

(a) Prove that $\{K_\delta\}_{\delta>0}$ is a family of good kernels.

(b) Assume in addition that φ is bounded and supported in a bounded set. Verify that $\{K_\delta\}_{\delta>0}$ is an approximation to the identity.

(c) Show that Theorem 2.3 (convergence in the L^1-norm) holds for good kernels as well.

2. Suppose $\{K_\delta\}$ is a family of kernels that satisfies:

(i) $|K_\delta(x)| \le A\delta^{-d}$ for all $\delta > 0$.

(ii) $|K_\delta(x)| \le A\delta/|x|^{d+1}$ for all $\delta > 0$.

(iii) $\int_{-\infty}^{\infty} K_\delta(x)\,dx = 0$ for all $\delta > 0$.

Thus K_δ satisfies conditions (i) and (ii) of approximations to the identity, but the average value of K_δ is 0 instead of 1. Show that if f is integrable on $\mathbb{R}^d$, then

$$(f * K_\delta)(x) \to 0 \qquad \text{for a.e. } x, \text{ as } \delta \to 0.$$

3. Suppose 0 is a point of (Lebesgue) density of the set $E \subset \mathbb{R}$. Show that for each of the individual conditions below there is an infinite sequence of points $x_n \in E$, with $x_n \neq 0$, and $x_n \to 0$ as $n \to \infty$.

(a) The sequence also satisfies $-x_n \in E$ for all n.

(b) In addition, $2x_n$ belongs to E for all n.

Generalize.

4. Prove that if f is integrable on $\mathbb{R}^d$, and f is not identically zero, then

$$f^*(x) \geq \frac{c}{|x|^d}, \qquad \text{for some } c > 0 \text{ and all } |x| \geq 1.$$

Conclude that f^* is not integrable on $\mathbb{R}^d$. Then, show that the weak type estimate

$$m(\{x : \ f^*(x) > \alpha\}) \leq c/\alpha$$

for all $\alpha > 0$ whenever $\int |f| = 1$, is best possible in the following sense: if f is supported in the unit ball with $\int |f| = 1$, then

$$m(\{x : \ f^*(x) > \alpha\}) \geq c'/\alpha$$

for some $c' > 0$ and all sufficiently small α.

[Hint: For the first part, use the fact that $\int_B |f| > 0$ for some ball B.]

5. Consider the function on $\mathbb{R}$ defined by

$$f(x) = \begin{cases} \dfrac{1}{|x|(\log 1/|x|)^2} & \text{if } |x| \le 1/2, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Verify that f is integrable.

(b) Establish the inequality

$$f^*(x) \ge \frac{c}{|x|(\log 1/|x|)} \qquad \text{for some } c > 0 \text{ and all } |x| \le 1/2,$$

to conclude that the maximal function f^* is not locally integrable.

6. In one dimension there is a version of the basic inequality (1) for the maximal function in the form of an identity. We define the "one-sided" maximal function

$$f_+^*(x) = \sup_{h>0} \frac{1}{h} \int_x^{x+h} |f(y)|\, dy.$$

If $E_\alpha^+ = \{x \in \mathbb{R} : \ f_+^*(x) > \alpha\}$, then

$$m(E_\alpha^+) = \frac{1}{\alpha} \int_{E_\alpha^+} |f(y)|\, dy.$$

[Hint: Apply Lemma 3.5 to $F(x) = \int_0^x |f(y)|\, dy - \alpha x$. Then E_α^+ is the union of disjoint intervals (a_k, b_k) with $\int_{a_k}^{b_k} |f(y)|\, dy = \alpha(a_k - b_k)$.]

7. Using Corollary 1.5, prove that if a measurable subset E of $[0,1]$ satisfies $m(E \cap I) \geq \alpha\, m(I)$ for some $\alpha > 0$ and all intervals I in $[0,1]$, then E has measure 1. See also Exercise 28 in Chapter 1.

8. Suppose A is a Lebesgue measurable set in $\mathbb{R}$ with $m(A) > 0$. Does there exist a sequence $\{s_n\}_{n=1}^{\infty}$ such that the complement of $\bigcup_{n=1}^{\infty}(A + s_n)$ in $\mathbb{R}$ has measure zero?

[Hint: For every $\epsilon > 0$, find an interval I_ϵ of length ℓ_ϵ such that $m(A \cap I_\epsilon) \geq (1 - \epsilon)m(I_\epsilon)$. Consider $\bigcup_{k=-\infty}^{\infty}(A + t_k)$, with $t_k = k\ell_\epsilon$. Then vary ϵ.]

9. Let F be a closed subset in $\mathbb{R}$, and $\delta(x)$ the distance from x to F, that is,

$$\delta(x) = d(x, F) = \inf\{|x - y| : \ y \in F\}.$$

Clearly, $\delta(x + y) \leq |y|$ whenever $x \in F$. Prove the more refined estimate

$$\delta(x + y) = o(|y|) \qquad \text{for a.e. } x \in F,$$

that is, $\delta(x + y)/|y| \to 0$ for a.e. $x \in F$.
[Hint: Assume that x is a point of density of F.]

10. Construct an increasing function on $\mathbb{R}$ whose set of discontinuities is precisely $\mathbb{Q}$.

11. If $a, b > 0$, let

$$f(x) = \begin{cases} x^a \sin(x^{-b}) & \text{for } 0 < x \leq 1, \\ 0 & \text{if } x = 0. \end{cases}$$

Prove that f is of bounded variation in $[0, 1]$ if and only if $a > b$. Then, by taking $a = b$, construct (for each $0 < \alpha < 1$) a function that satisfies the Lipschitz condition of exponent α

$$|f(x) - f(y)| \leq A|x - y|^\alpha$$

but which is not of bounded variation.

[Hint: Note that if $h > 0$, the difference $|f(x + h) - f(x)|$ can be estimated by $C(x + h)^a$, or $C'h/x$ by the mean value theorem. Then, consider two cases, whether $x^{a+1} \geq h$ or $x^{a+1} < h$. What is the relationship between α and a?]

12. Consider the function $F(x) = x^2 \sin(1/x^2)$, $x \neq 0$, with $F(0) = 0$. Show that $F'(x)$ exists for every x, but F' is not integrable on $[-1, 1]$.

13. Show directly from the definition that the Cantor-Lebesgue function is not absolutely continuous.

14. The following measurability issues arose in the discussion of differentiability of functions.

(a) Suppose F is continuous on $[a, b]$. Show that

$$D^+(F)(x) = \limsup_{\substack{h \to 0 \\ h > 0}} \frac{F(x+h) - F(x)}{h}$$

is measurable.

(b) Suppose $J(x) = \sum_{n=1}^{\infty} \alpha_n j_n(x)$ is a jump function as in Section 3.3. Show that

$$\limsup_{h \to 0} \frac{J(x+h) - J(x)}{h}$$

is measurable.

[Hint: For (a), the continuity of F allows one to restrict to countably many h in taking the limsup. For (b), given $k > m$, let $F_{k,m}^N = \sup_{1/k \le |h| \le 1/m} \left| \frac{J_N(x+h) - J_N(x)}{h} \right|$, where $J_N(x) = \sum_{n=1}^{N} \alpha_n j_n(x)$. Note that each $F_{k,m}^N$ is measurable. Then, successively, let $N \to \infty$, $k \to \infty$, and finally $m \to \infty$.]

15. Suppose F is of bounded variation and continuous. Prove that $F = F_1 - F_2$, where both F_1 and F_2 are monotonic and continuous.

16. Show that if F is of bounded variation in $[a, b]$, then:

(a) $\int_a^b |F'(x)|\,dx \leq T_F(a, b)$.

(b) $\int_a^b |F'(x)|\,dx = T_F(a, b)$ if and only if F is absolutely continuous.

As a result of (b), the formula $L = \int_a^b |z'(t)|\,dt$ for the length of a rectifiable curve parametrized by z holds if and only if z is absolutely continuous.

17. Prove that if $\{K_\epsilon\}_{\epsilon>0}$ is a family of approximations to the identity, then

$$\sup_{\epsilon>0} |(f * K_\epsilon)(x)| \leq cf^*(x)$$

for some constant $c > 0$ and all integrable f.

18. Verify the agreement between the two definitions given for the Cantor-Lebesgue function in Exercise 2, Chapter 1 and in Section 3.1 of this chapter.

19. Show that if $f : \mathbb{R} \to \mathbb{R}$ is absolutely continuous, then

(a) f maps sets of measure zero to sets of measure zero.

(b) f maps measurable sets to measurable sets.

20. This exercise deals with functions F that are absolutely continuous on $[a, b]$ and are increasing. Let $A = F(a)$ and $B = F(b)$.

(a) There exists such an F that is in addition strictly increasing, but such that $F'(x) = 0$ on a set of positive measure.

(b) The F in (a) can be chosen so that there is a measurable subset $E \subset [A, B]$, $m(E) = 0$, so that $F^{-1}(E)$ is not measurable.

(c) Prove, however, that for any increasing absolutely continuous F, and E a measurable subset of $[A, B]$, the set $F^{-1}(E) \cap \{F'(x) > 0\}$ is measurable.

[Hint: (a) Let $F(x) = \int_a^x \chi_K(x)\, dx$, where K is the complement of a Cantor-like set C of positive measure. For (b), note that $F(C)$ is a set of measure zero. Finally, for (c) prove first that $m(\mathcal{O}) = \int_{F^{-1}(\mathcal{O})} F'(x)\, dx$ for any open set $\mathcal{O}$.]

21. Let F be absolutely continuous and increasing on $[a, b]$ with $F(a) = A$ and $F(b) = B$. Suppose f is any measurable function on $[A, B]$.

(a) Show that $f(F(x))F'(x)$ is measurable on $[a, b]$. Note: $f(F(x))$ need not be measurable by Exercise 20 (b).

(b) Prove the change of variable formula: If f is integrable on $[A, B]$, then so is $f(F(x))F'(x)$, and

$$\int_A^B f(y)\,dy = \int_a^b f(F(x))F'(x)\,dx.$$

[Hint: Start with the identity $m(\mathcal{O}) = \int_{F^{-1}(\mathcal{O})} F'(x)\,dx$ used in (c) of Exercise 20 above.]

22. Suppose that F and G are absolutely continuous on $[a, b]$. Show that their product FG is also absolutely continuous. This has the following consequences.

(a) Whenever F and G are absolutely continuous in $[a, b]$,

$$\int_a^b F'(x)G(x)\,dx = -\int_a^b F(x)G'(x)\,dx + [F(x)G(x)]_a^b.$$

(b) Let F be absolutely continuous in $[-\pi, \pi]$ with $F(\pi) = F(-\pi)$. Show that if

$$a_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} F(x)e^{-inx}\,dx,$$

such that $F(x) \sim \sum a_n e^{inx}$, then

$$F'(x) \sim \sum ina_n e^{inx}.$$

(c) What happens if $F(-\pi) \neq F(\pi)$? [Hint: Consider $F(x) = x$.]

23. Let F be continuous on $[a, b]$. Show the following.

(a) Suppose $(D^+F)(x) \geq 0$ for every $x \in [a, b]$. Then F is increasing on $[a, b]$.

(b) If $F'(x)$ exists for every $x \in (a, b)$ and $|F'(x)| \leq M$, then $|F(x) - F(y)| \leq M|x - y|$ and F is absolutely continuous.

[Hint: For (a) it suffices to show that $F(b) - F(a) \geq 0$. Assume otherwise. Hence with $G_\epsilon(x) = F(x) - F(a) + \epsilon(x - a)$, for sufficiently small $\epsilon > 0$ we have $G_\epsilon(a) = 0$, but $G_\epsilon(b) < 0$. Now let $x_0 \in [a, b)$ be the greatest value of x_0 such that $G_\epsilon(x_0) \geq 0$. However, $(D^+G_\epsilon)(x_0) > 0$.]

24. Suppose F is an increasing function on $[a, b]$.

(a) Prove that we can write

$$F = F_A + F_C + F_J,$$

where each of the functions F_A, F_C, and F_J is increasing and:

(i) F_A is absolutely continuous.

(ii) F_C is continuous, but $F_C'(x) = 0$ for a.e. x.

(iii) F_J is a jump function.

(b) Moreover, each component F_A, F_C, F_J is uniquely determined up to an additive constant.

The above is the **Lebesgue decomposition** of F. There is a corresponding decomposition for any F of bounded variation.

25. The following shows the necessity of allowing for general exceptional sets of measure zero in the differentiation Theorems 1.4, 3.4, and 3.11. Let E be any set of measure zero in $\mathbb{R}^d$. Show that:

(a) There exists a non-negative integrable f in $\mathbb{R}^d$, so that

$$\liminf_{\substack{m(B)\to 0 \\ x\in B}} \frac{1}{m(B)}\int_B f(y)\,dy = \infty \qquad \text{for each } x \in E.$$

(b) When $d = 1$ this may be restated as follows. There is an increasing absolutely continuous function F so that

$$D_+(F)(x) = D_-(F)(x) = \infty, \qquad \text{for each } x \in E.$$

[Hint: Find open sets $\mathcal{O}_n \supset E$, with $m(\mathcal{O}_n) < 2^{-n}$, and let $f(x) = \sum_{n=1}^{\infty} \chi_{\mathcal{O}_n}(x)$.]

26. An alternative way of defining the exterior measure $m_*(E)$ of an arbitrary set E, as given in Section 2 of Chapter 1, is to replace the coverings of E by cubes with coverings by balls. That is, suppose we define $m_*^{\mathcal{B}}(E)$ as $\inf \sum_{j=1}^{\infty} m(B_j)$, where the infimum is taken over all coverings $E \subset \bigcup_{j=1}^{\infty} B_j$ by open balls. Then $m_*(E) = m_*^{\mathcal{B}}(E)$. (Observe that this result leads to an alternate proof that the Lebesgue measure is invariant under rotations.)

Clearly $m_*(E) \leq m_*^{\mathcal{B}}(E)$. Prove the reverse inequality by showing the following. For any $\epsilon > 0$, there is a collection of balls $\{B_j\}$ such that $E \subset \bigcup_j B_j$ while $\sum_j m(B_j) \leq m_*(E) + \epsilon$. Note also that for any preassigned δ, we can choose the balls to have diameter $< \delta$.

[Hint: Assume first that E is measurable, and pick $\mathcal{O}$ open so that $\mathcal{O} \supset E$ and $m(\mathcal{O} - E) < \epsilon'$. Next, using Corollary 3.10, find balls $B_1, \ldots, B_N$ such that $\sum_{j=1}^{N} m(B_j) \leq m(E) + 2\epsilon'$ and $m(E - \bigcup_{j=1}^{N} B_j) \leq 3\epsilon'$. Finally, cover $E - \bigcup_{j=1}^{N} B_j$ by a union of cubes, the sum of whose measures is $\leq 4\epsilon'$, and replace these cubes by balls that contain them. For the general E, begin by applying the above when E is a cube.]

27. A rectifiable curve has a tangent line at almost all points of the curve. Make this statement precise.

28. A curve in $\mathbb{R}^d$ is a continuous map $t \mapsto z(t)$ of an interval $[a, b]$ into $\mathbb{R}^d$.

(a) State and prove the analogues of the conditions dealing with the rectifiability of curves and their length that are given in Theorems 3.1, 4.1, and 4.3.

(b) Define the (one-dimensional) Minkowski content $\mathcal{M}(K)$ of a compact set in $\mathbb{R}^d$ as the limit (if it exists) of

$$\frac{m(K^\delta)}{m_{d-1}(B(\delta))} \qquad \text{as } \delta \to 0,$$

where $m_{d-1}(B(\delta))$ is the measure (in $\mathbb{R}^{d-1}$) of the ball defined by $B(\delta) = \{x \in \mathbb{R}^{d-1},\ |x| < \delta\}$. State and prove analogues of Propositions 4.5 and 4.7 for curves in $\mathbb{R}^d$.

29. Let $\Gamma = \{z(t),\ a \le t \le b\}$ be a curve, and suppose it satisfies a Lipschitz condition with exponent α, $1/2 \le \alpha \le 1$, that is,

$$|z(t) - z(t')| \le A|t - t'|^\alpha \qquad \text{for all } t, t' \in [a, b].$$

Show that $m(\Gamma^\delta) = O(\delta^{2-1/\alpha})$ for $0 < \delta \le 1$.

30. A bounded function F is said to be of bounded variation on $\mathbb{R}$ if F is of bounded variation on any finite sub-interval $[a, b]$, and $\sup_{a,b} T_F(a, b) < \infty$.

Prove that such an F enjoys the following two properties:

(a) $\int_{\mathbb{R}} |F(x + h) - F(x)|\, dx \leq A|h|$, for some constant A and all $h \in \mathbb{R}$.

(b) $|\int_{\mathbb{R}} F(x)\varphi'(x)\, dx| \leq A$, where φ ranges over all C^1 functions of bounded support with $\sup_{x\in\mathbb{R}} |\varphi(x)| \leq 1$.

For the converse, and analogues in $\mathbb{R}^d$, see Problem 6^* below.

[Hint: For (a), write $F = F_1 - F_2$, where F_j are monotonic and bounded. For (b), deduce this from (a).]

31. Let F be the Cantor-Lebesgue function described in Section 3.1. Consider the curve that is the graph of F, that is, the curve given by $x(t) = t$ and $y(t) = F(t)$ with $0 \leq t \leq 1$. Prove that the length $L(\overline{x})$ of the segment $0 \leq t \leq \overline{x}$ of the curve is given by $L(\overline{x}) = \overline{x} + F(\overline{x})$. Hence the total length of the curve is 2.

32. Let $f : \mathbb{R} \to \mathbb{R}$. Prove that f satisfies the Lipschitz condition

$$|f(x) - f(y)| \leq M|x - y|$$

for some M and all $x, y \in \mathbb{R}$, if and only if f satisfies the following two properties:

(i) f is absolutely continuous.

(ii) $|f'(x)| \leq M$ for a.e. x.

Chapter 4. Hilbert Spaces: An Introduction

1. Show that properties (i) and (ii) in the definition of a Hilbert space (Section 2) imply property (iii): the Cauchy-Schwarz inequality $|(f, g)| \leq \|f\| \cdot \|g\|$ and the triangle inequality $\|f + g\| \leq \|f\| + \|g\|$.

[Hint: For the first inequality, consider $(f + \lambda g, f + \lambda g)$ as a positive quadratic function of λ. For the second, write $\|f + g\|^2$ as $(f + g, f + g)$.]

2. In the case of equality in the Cauchy-Schwarz inequality we have the following. If $|(f,g)| = \|f\|\,\|g\|$ and $g \neq 0$, then $f = cg$ for some scalar c.

[Hint: Assume $\|f\| = \|g\| = 1$ and $(f,g) = 1$. Then $f - g$ and g are orthogonal, while $f = f - g + g$. Thus $\|f\|^2 = \|f - g\|^2 + \|g\|^2$.]

3. Note that $\|f+g\|^2 = \|f\|^2 + \|g\|^2 + 2\mathrm{Re}(f,g)$ for any pair of elements in a Hilbert space $\mathcal{H}$. As a result, verify the identity $\|f+g\|^2 + \|f-g\|^2 = 2(\|f\|^2 + \|g\|^2)$.

4. Prove from the definition that $\ell^2(\mathbb{Z})$ is complete and separable.

5. Establish the following relations between $L^2(\mathbb{R}^d)$ and $L^1(\mathbb{R}^d)$:

(a) Neither the inclusion $L^2(\mathbb{R}^d) \subset L^1(\mathbb{R}^d)$ nor the inclusion $L^1(\mathbb{R}^d) \subset L^2(\mathbb{R}^d)$ is valid.

(b) Note, however, that if f is supported on a set E of finite measure and if $f \in L^2(\mathbb{R}^d)$, applying the Cauchy-Schwarz inequality to $f\chi_E$ gives $f \in L^1(\mathbb{R}^d)$, and

$$\|f\|_{L^1(\mathbb{R}^d)} \leq m(E)^{1/2}\|f\|_{L^2(\mathbb{R}^d)}.$$

(c) If f is bounded ($|f(x)| \leq M$), and $f \in L^1(\mathbb{R}^d)$, then $f \in L^2(\mathbb{R}^d)$ with

$$\|f\|_{L^2(\mathbb{R}^d)} \leq M^{1/2}\|f\|^{1/2}_{L^1(\mathbb{R}^d)}.$$

[Hint: For (a) consider $f(x) = |x|^{-\alpha}$, when $|x| \leq 1$ or when $|x| > 1$.]

6. Prove that the following are dense subspaces of $L^2(\mathbb{R}^d)$.

(a) The simple functions.

(b) The continuous functions of compact support.

7. Suppose $\{\varphi_k\}_{k=1}^{\infty}$ is an orthonormal basis for $L^2(\mathbb{R}^d)$. Prove that the collection $\{\varphi_{k,j}\}_{1\leq k,j<\infty}$ with $\varphi_{k,j}(x,y) = \varphi_k(x)\varphi_j(y)$ is an orthonormal basis of $L^2(\mathbb{R}^d \times \mathbb{R}^d)$.

[Hint: First verify that the $\{\varphi_{k,j}\}$ are orthonormal, by Fubini's theorem. Next, for each j consider $F_j(x) = \int_{\mathbb{R}^d} F(x,y)\overline{\varphi_j(y)}\, dy$. If one assumes that $(F, \varphi_{k,j}) = 0$ for all j, then $\int F_j(x)\overline{\varphi_k(x)}\, dx = 0$.]

8. Let $\eta(t)$ be a fixed strictly positive continuous function on $[a, b]$. Define $\mathcal{H}_\eta = L^2([a, b], \eta)$ to be the space of all measurable functions f on $[a, b]$ such that

$$\int_a^b |f(t)|^2 \eta(t)\, dt < \infty.$$

Define the inner product on $\mathcal{H}_\eta$ by

$$(f, g)_\eta = \int_a^b f(t)\overline{g(t)}\eta(t)\, dt.$$

(a) Show that $\mathcal{H}_\eta$ is a Hilbert space, and that the mapping $U : f \mapsto \eta^{1/2} f$ gives a unitary correspondence between $\mathcal{H}_\eta$ and the usual space $L^2([a, b])$.

(b) Generalize this to the case when η is not necessarily continuous.

9. Let $\mathcal{H}_1 = L^2([-\pi, \pi])$ be the Hilbert space of functions $F(e^{i\theta})$ on the unit circle with inner product $(F, G) = \frac{1}{2\pi} \int_{-\pi}^{\pi} F(e^{i\theta})\overline{G(e^{i\theta})}\, d\theta$. Let $\mathcal{H}_2$ be the space $L^2(\mathbb{R})$. Using the mapping

$$x \mapsto \frac{i - x}{i + x}$$

of $\mathbb{R}$ to the unit circle, show that:

(a) The correspondence $U : F \to f$, with

$$f(x) = \frac{1}{\pi^{1/2}(i + x)} F\left(\frac{i - x}{i + x}\right)$$

gives a unitary mapping of $\mathcal{H}_1$ to $\mathcal{H}_2$.

(b) As a result,

$$\left\{ \frac{1}{\pi^{1/2}} \left(\frac{i - x}{i + x}\right)^n \frac{1}{i + x} \right\}_{n=-\infty}^{\infty}$$

is an orthonormal basis of $L^2(\mathbb{R})$.

10. Let $\mathcal{S}$ denote a subspace of a Hilbert space $\mathcal{H}$. Prove that $(\mathcal{S}^{\perp})^{\perp}$ is the smallest closed subspace of $\mathcal{H}$ that contains $\mathcal{S}$.

11. Let P be the orthogonal projection associated with a closed subspace $\mathcal{S}$ in a Hilbert space $\mathcal{H}$, that is,

$$P(f) = f \text{ if } f \in \mathcal{S} \qquad \text{and} \qquad P(f) = 0 \text{ if } f \in \mathcal{S}^{\perp}.$$

(a) Show that $P^2 = P$ and $P^* = P$.

(b) Conversely, if P is any bounded operator satisfying $P^2 = P$ and $P^* = P$, prove that P is the orthogonal projection for some closed subspace of $\mathcal{H}$.

(c) Using P, prove that if $\mathcal{S}$ is a closed subspace of a separable Hilbert space, then $\mathcal{S}$ is also a separable Hilbert space.

12. Let E be a measurable subset of $\mathbb{R}^d$, and suppose $\mathcal{S}$ is the subspace of $L^2(\mathbb{R}^d)$ of functions that vanish for a.e. $x \notin E$. Show that the orthogonal projection P on $\mathcal{S}$ is given by $P(f) = \chi_E \cdot f$, where χ_E is the characteristic function of E.

13. Suppose P_1 and P_2 are a pair of orthogonal projections on S_1 and S_2, respectively. Then P_1P_2 is an orthogonal projection if and only if P_1 and P_2 commute, that is, $P_1P_2 = P_2P_1$. In this case, P_1P_2 projects onto $S_1 \cap S_2$.

14. Suppose $\mathcal{H}$ and $\mathcal{H}'$ are two completions of a pre-Hilbert space $\mathcal{H}_0$. Show that there is a unitary mapping from $\mathcal{H}$ to $\mathcal{H}'$ that is the identity on $\mathcal{H}_0$.

[Hint: If $f \in \mathcal{H}$, pick a Cauchy sequence $\{f_n\}$ in $\mathcal{H}_0$ that converges to f in $\mathcal{H}$. This sequence will also converge to an element f' in $\mathcal{H}'$. The mapping $f \mapsto f'$ gives the required unitary mapping.]

15. Let T be any linear transformation from $\mathcal{H}_1$ to $\mathcal{H}_2$. If we suppose that $\mathcal{H}_1$ is finite-dimensional, then T is automatically bounded. (If $\mathcal{H}_1$ is not assumed to be finite-dimensional this may fail; see Problem 1 below.)

16. Let $F_0(z) = 1/(1-z)^i$.

(a) Verify that $|F_0(z)| \leq e^{\pi/2}$ in the unit disc, but that $\lim_{r\to 1} F_0(r)$ does not exist.

[Hint: Note that $|F_0(r)| = 1$ and $F_0(r)$ oscillates between ± 1 infinitely often as $r \to 1$.]

(b) Let $\{\alpha_n\}_{n=1}^{\infty}$ be an enumeration of the rationals, and let

$$F(z) = \sum_{j=1}^{\infty} \delta^j F_0(ze^{-i\alpha_j}),$$

where δ is sufficiently small. Show that $\lim_{r\to 1} F(re^{i\theta})$ fails to exist whenever $\theta = \alpha_j$, and hence F fails to have a radial limit for a dense set of points on the unit circle.

17. Fatou's theorem can be generalized by allowing a point to approach the boundary in larger regions, as follows.

For each $0 < s < 1$ and point z on the unit circle, consider the region $\Gamma_s(z)$ defined as the smallest closed convex set that contains z and the closed disc $D_s(0)$. In other words, $\Gamma_s(z)$ consists of all lines joining z with points in $D_s(0)$. Near the point z, the region $\Gamma_s(z)$ looks like a triangle. See Figure 2.

We say that a function F defined in the open unit disc has a **non-tangential limit** at a point z on the circle, if for every $0 < s < 1$, the limit

$$\lim_{\substack{w \to z \\ w \in \Gamma_s(z)}} F(w)$$

exists.

Prove that if F is holomorphic and bounded on the open unit disc, then F has a non-tangential limit for almost every point on the unit circle.
[Hint: Show that the Poisson integral of a function f has non-tangential limits at every point of the Lebesgue set of f.]

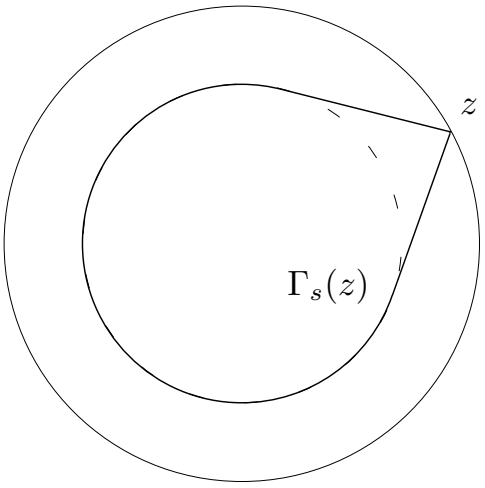


Figure 2. The region $\Gamma_s(z)$

18. Let $\mathcal{H}$ denote a Hilbert space, and $\mathcal{L}(\mathcal{H})$ the vector space of all bounded linear operators on $\mathcal{H}$. Given $T \in \mathcal{L}(\mathcal{H})$, we define the operator norm

$$\|T\| = \inf\{B : \|Tv\| \leq B\|v\|, \quad \text{for all } v \in \mathcal{H}\}.$$

(a) Show that $\|T_1 + T_2\| \leq \|T_1\| + \|T_2\|$ whenever $T_1, T_2 \in \mathcal{L}(\mathcal{H})$.

(b) Prove that

$$d(T_1, T_2) = \|T_1 - T_2\|$$

defines a metric on $\mathcal{L}(\mathcal{H})$.

(c) Show that $\mathcal{L}(\mathcal{H})$ is complete in the metric d.

19. If T is a bounded linear operator on a Hilbert space, prove that

$$\|TT^*\| = \|T^*T\| = \|T\|^2 = \|T^*\|^2.$$

20. Suppose $\mathcal{H}$ is an infinite-dimensional Hilbert space. We have seen an example of a sequence $\{f_n\}$ in $\mathcal{H}$ with $\|f_n\| = 1$ for all n, but for which no subsequence of $\{f_n\}$ converges in $\mathcal{H}$. However, show that for any sequence $\{f_n\}$ in $\mathcal{H}$ with $\|f_n\| = 1$ for all n, there exist $f \in \mathcal{H}$ and a subsequence $\{f_{n_k}\}$ such that for all $g \in \mathcal{H}$, one has

$$\lim_{k\to\infty} (f_{n_k}, g) = (f, g).$$

One says that $\{f_{n_k}\}$ **converges weakly** to f.

[Hint: Let g run through a basis for $\mathcal{H}$, and use a diagonalization argument. One can then define f by giving its series expansion with respect to the chosen basis.]

21. There are several senses in which a sequence of bounded operators $\{T_n\}$ can converge to a bounded operator T (in a Hilbert space $\mathcal{H}$). First, there is convergence in the norm, that is, $\|T_n - T\| \to 0$, as $n \to \infty$. Next, there is a weaker convergence, which happens to be called **strong convergence**, that requires that $T_n f \to Tf$, as $n \to \infty$, for every vector $f \in \mathcal{H}$. Finally, there is **weak convergence** (see also Exercise 20) that requires $(T_n f, g) \to (Tf, g)$ for every pair of vectors $f, g \in \mathcal{H}$.

(a) Show by examples that weak convergence does not imply strong convergence, nor does strong convergence imply convergence in the norm.

(b) Show that for any bounded operator T there is a sequence $\{T_n\}$ of bounded operators of finite rank so that $T_n \to T$ strongly as $n \to \infty$.

22. An operator T is an **isometry** if $\|Tf\| = \|f\|$ for all $f \in \mathcal{H}$.

(a) Show that if T is an isometry, then $(Tf, Tg) = (f, g)$ for every $f, g \in \mathcal{H}$. Prove as a result that $T^*T = I$.

(b) If T is an isometry and T is surjective, then T is unitary and $TT^* = I$.

(c) Give an example of an isometry that is not unitary.

(d) Show that if T^*T is unitary then T is an isometry.

[Hint: Use the fact that $(Tf, Tf) = (f, f)$ for f replaced by $f \pm g$ and $f \pm ig$.]

23. Suppose $\{T_k\}$ is a collection of bounded operators on a Hilbert space $\mathcal{H}$, with $\|T_k\| \leq 1$ for all k. Suppose also that

$$T_k T_j^* = T_k^* T_j = 0 \qquad \text{for all } k \neq j.$$

Let $S_N = \sum_{k=-N}^{N} T_k$.

Show that $S_N(f)$ converges as $N \to \infty$, for every $f \in \mathcal{H}$. If $T(f)$ denotes the limit, prove that $\|T\| \leq 1$.

A generalization is given in Problem 8^* below.

[Hint: Consider first the case when only finitely many of the T_k are non-zero, and note that the ranges of the T_k are mutually orthogonal.]

24. Let $\{e_k\}_{k=1}^{\infty}$ denote an orthonormal set in a Hilbert space $\mathcal{H}$. If $\{c_k\}_{k=1}^{\infty}$ is a sequence of positive real numbers such that $\sum c_k^2 < \infty$, then the set

$$A = \{\sum_{k=1}^{\infty} a_k e_k : \ |a_k| \leq c_k\}$$

is compact in $\mathcal{H}$.

25. Suppose T is a bounded operator that is diagonal with respect to a basis $\{\varphi_k\}$, with $T\varphi_k = \lambda_k \varphi_k$. Then T is compact if and only if $\lambda_k \to 0$.

[Hint: If $\lambda_k \to 0$, then note that $\|P_n T - T\| \to 0$, where P_n is the orthogonal projection on the subspace spanned by $\varphi_1, \varphi_2, \ldots, \varphi_n$.]

26. Suppose w is a measurable function on $\mathbb{R}^d$ with $0 < w(x) < \infty$ for a.e. x, and K is a measurable function on $\mathbb{R}^{2d}$ that satisfies:

(i) $\displaystyle\int_{\mathbb{R}^d} |K(x,y)|w(y)\,dy \le Aw(x)$ for almost every $x \in \mathbb{R}^d$, and

(ii) $\displaystyle\int_{\mathbb{R}^d} |K(x,y)|w(x)\,dx \le Aw(y)$ for almost every $y \in \mathbb{R}^d$.

Prove that the integral operator defined by

$$Tf(x) = \int_{\mathbb{R}^d} K(x,y)f(y)\,dy, \qquad x \in \mathbb{R}^d$$

is bounded on $L^2(\mathbb{R}^d)$ with $\|T\| \le A$.

Note as a special case that if $\int |K(x,y)|\,dy \le A$ for all x, and $\int |K(x,y)|\,dx \le A$ for all y, then $\|T\| \le A$.

[Hint: Show that if $f \in L^2(\mathbb{R}^d)$, then

$$\int |K(x,y)|\,|f(y)|\,dy \le A^{1/2}w(x)^{1/2}\left[\int |K(x,y)|\,|f(y)|^2 w(y)^{-1}\,dy\right]^{1/2}.]$$

27. Prove that the operator

$$Tf(x) = \frac{1}{\pi} \int_0^\infty \frac{f(y)}{x+y}\, dy$$

is bounded on $L^2(0,\infty)$ with norm $\|T\| \leq 1$.
[Hint: Use Exercise 26 with an appropriate w.]

28. Suppose $\mathcal{H} = L^2(B)$, where B is the unit ball in $\mathbb{R}^d$. Let $K(x, y)$ be a measurable function on $B \times B$ that satisfies $|K(x, y)| \leq A|x - y|^{-d+\alpha}$ for some $\alpha > 0$, whenever $x, y \in B$. Define

$$Tf(x) = \int_B K(x, y) f(y) dy.$$

(a) Prove that T is a bounded operator on $\mathcal{H}$.

(b) Prove that T is compact.

(c) Note that T is a Hilbert-Schmidt operator if and only if $\alpha > d/2$.

[Hint: For (b), consider the operators T_n associated with the truncated kernels $K_n(x, y) = K(x, y)$ if $|x - y| \geq 1/n$ and 0 otherwise. Show that each T_n is compact, and that $\|T_n - T\| \to 0$ as $n \to \infty$.]

29. Let T be a compact operator on a Hilbert space $\mathcal{H}$, and assume $\lambda \neq 0$.

(a) Show that the range of $\lambda I - T$ defined by

$$\{g \in \mathcal{H} : g = (\lambda I - T)f, \text{ for some } f \in \mathcal{H}\}$$

is closed. [Hint: Suppose $g_j \to g$, where $g_j = (\lambda I - T)f_j$. Let V_λ denote the eigenspace of T corresponding to λ, that is, the kernel of $\lambda I - T$. Why can one assume that $f_j \in V_\lambda^\perp$? Under this assumption prove that $\{f_j\}$ is a bounded sequence.]

(b) Show by example that this may fail when $\lambda = 0$.

(c) Show that the range of $\lambda I - T$ is all of $\mathcal{H}$ if and only if the null-space of $\overline{\lambda} I - T^*$ is trivial.

30. Let $\mathcal{H} = L^2([-\pi, \pi])$ with $[-\pi, \pi]$ identified as the unit circle. Fix a bounded sequence $\{\lambda_n\}_{n=-\infty}^{\infty}$ of complex numbers, and define an operator Tf by

$$Tf(x) \sim \sum_{n=-\infty}^{\infty} \lambda_n a_n e^{inx} \qquad \text{whenever} \qquad f(x) \sim \sum_{n=-\infty}^{\infty} a_n e^{inx}.$$

Such an operator is called a **Fourier multiplier operator**, and the sequence $\{\lambda_n\}$ is called the **multiplier sequence**.

(a) Show that T is a bounded operator on $\mathcal{H}$ and $\|T\| = \sup_n |\lambda_n|$.

(b) Verify that T commutes with translations, that is, if we define $\tau_h(x) = f(x - h)$ then

$$T \circ \tau_h = \tau_h \circ T \qquad \text{for every } h \in \mathbb{R}.$$

(c) Conversely, prove that if T is any bounded operator on $\mathcal{H}$ that commutes with translations, then T is a Fourier multiplier operator. [Hint: Consider $T(e^{inx})$.]

31. Consider a version of the sawtooth function defined on $[-\pi, \pi)$ by[5]

$$K(x) = i(sgn(x)\pi - x),$$

and extended to $\mathbb{R}$ with period 2π. Suppose $f \in L^1([-\pi, \pi])$ is extended to $\mathbb{R}$ with period 2π, and define

$$\begin{aligned} Tf(x) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} K(x-y) f(y)\, dy \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} K(y) f(x-y)\, dy. \end{aligned}$$

(a) Show that $F(x) = Tf(x)$ is absolutely continuous, and if $\int_{-\pi}^{\pi} f(y)dy = 0$, then $F'(x) = if(x)$ a.e. x.

(b) Show that the mapping $f \mapsto Tf$ is compact and symmetric on $L^2([-\pi, \pi])$.

(c) Prove that $\varphi(x) \in L^2([-\pi, \pi])$ is an eigenfunction for T if and only if $\varphi(x)$ is (up to a constant multiple) equal to e^{inx} for some integer $n \neq 0$ with eigenvalue $1/n$, or $\varphi(x) = 1$ with eigenvalue 0.

(d) Show as a result that $\{e^{inx}\}_{n\in\mathbb{Z}}$ is an orthonormal basis of $L^2([-\pi, \pi])$.

Note that in Book I, Chapter 2, Exercise 8, it is shown that the Fourier series of K is

$$K(x) \sim \sum_{n\neq 0} \frac{e^{inx}}{n}.$$

[5]The symbol $sgn(x)$ denotes the sign function: it equals 1 or -1 if x is positive or negative respectively, and 0 if $x = 0$.

32. Consider the operator $T : L^2([0,1]) \to L^2([0,1])$ defined by

$$T(f)(t) = tf(t).$$

(a) Prove that T is a bounded linear operator with $T = T^*$, but that T is not compact.

(b) However, show that T has no eigenvectors.

33. Let $\mathcal{H}$ be a Hilbert space with basis $\{\varphi_k\}_{k=1}^{\infty}$. Verify that the operator T defined by

$$T(\varphi_k) = \frac{1}{k}\,\varphi_{k+1}$$

is compact, but has no eigenvectors.

34. Let K be a Hilbert-Schmidt kernel which is real and symmetric. Then, as we saw, the operator T whose kernel is K is compact and symmetric. Let $\{\varphi_k(x)\}$ be the eigenvectors (with eigenvalues λ_k) that diagonalize T. Then:

(a) $\sum_k |\lambda_k|^2 < \infty$.

(b) $K(x,y) \sim \sum \lambda_k \varphi_k(x)\varphi_k(y)$ is the expansion of K in the basis $\{\varphi_k(x)\varphi_k(y)\}$.

(c) Suppose T is a compact operator which is symmetric. Then T is of Hilbert-Schmidt type if and only if $\sum_n |\lambda_n|^2 < \infty$, where $\{\lambda_n\}$ are the eigenvalues of T counted according to their multiplicities.

35. Let $\mathcal{H}$ be a Hilbert space. Prove the following variants of the spectral theorem.

(a) If T_1 and T_2 are two linear symmetric and compact operators on $\mathcal{H}$ that commute (that is, $T_1T_2 = T_2T_1$), show that they can be diagonalized simultaneously. In other words, there exists an orthonormal basis for $\mathcal{H}$ which consists of eigenvectors for both T_1 and T_2.

(b) A linear operator on $\mathcal{H}$ is **normal** if $TT^* = T^*T$. Prove that if T is normal and compact, then T can be diagonalized.

[Hint: Write $T = T_1 + iT_2$ where T_1 and T_2 are symmetric, compact and commute.]

(c) If U is unitary, and $U = \lambda I - T$, where T is compact, then U can be diagonalized.

Chapter 5. Hilbert Spaces: Several Examples

1. Suppose $f \in L^2(\mathbb{R}^d)$ and $k \in L^1(\mathbb{R}^d)$.

(a) Show that $(f * k)(x) = \int f(x-y)k(y)\,dy$ converges for a.e. x.

(b) Prove that $\|f * k\|_{L^2(\mathbb{R}^d)} \leq \|f\|_{L^2(\mathbb{R}^d)}\|k\|_{L^1(\mathbb{R}^d)}$.

(c) Establish $\widehat{(f * k)}(\xi) = \hat{k}(\xi)\hat{f}(\xi)$ for a.e. ξ.

(d) The operator $Tf = f * k$ is a Fourier multiplier operator with multiplier $m(\xi) = \hat{k}(\xi)$.

[Hint: See Exercise 21 in Chapter 2.]

2. Consider the **Mellin transform** defined initially for continuous functions f of compact support in $\mathbb{R}^+ = \{t \in \mathbb{R} : \ t > 0\}$ and $x \in \mathbb{R}$ by

$$\mathcal{M}f(x) = \int_0^\infty f(t)t^{ix-1}dt.$$

Prove that $(2\pi)^{-1/2}\mathcal{M}$ extends to a unitary operator from $L^2(\mathbb{R}^+, dt/t)$ to $L^2(\mathbb{R})$. The Mellin transform serves on $\mathbb{R}^+$, with its multiplicative structure, the same purpose as the Fourier transform on $\mathbb{R}$, with its additive structure.

3. Let $F(z)$ be a bounded holomorphic function in the half-plane. Show in two ways that $\lim_{y\to 0} F(x+iy)$ exists for a.e. x.

(a) By using the fact that $F(z)/(z+i)$ is in $H^2(\mathbb{R}^2_+)$.

(b) By noting that $G(z) = F\left(i\frac{1-z}{1+z}\right)$ is a bounded holomorphic function in the unit disc, and using Exercise 17 in the previous chapter.

4. Consider $F(z) = e^{i/z}/(z+i)$ in the upper half-plane. Note that $F(x+iy) \in L^2(\mathbb{R})$, for each $y > 0$ and $y = 0$. Observe also that $F(z) \to 0$ as $|z| \to 0$. However, $F \notin H^2(\mathbb{R}^2_+)$. Why?

5. For $a < b$, let $S_{a,b}$ denote the strip $\{z = x + iy, \quad a < y < b\}$. Define $H^2(S_{a,b})$ to consist of the holomorphic functions F in $S_{a,b}$ so that

$$\|F\|^2_{H^2(S_{a,b})} = \sup_{a<y<b} \int_{\mathbb{R}^2} |F(x+iy)|^2 \, dx < \infty.$$

Define $H^2(S_{a,\infty})$ and $H^2(S_{-\infty,b})$ to be the obvious variants of the Hardy spaces for the half-planes $\{z = x + iy, \ y > a\}$ and $\{z = x + iy, \ y < b\}$, respectively.

(a) Show that $F \in H^2(S_{a,b})$ if and only if F can be written as

$$F(z) = \int_{\mathbb{R}} f(\xi) e^{-2\pi i z \xi} \, d\xi,$$

with $\int_{\mathbb{R}} |f(\xi)|^2 (e^{4\pi a \xi} + e^{4\pi b \xi}) \, d\xi < \infty$.

(b) Prove that every $F \in H^2(S_{a,b})$ can be decomposed as $F = G_1 + G_2$, where $G_{\in} H^2(S_{a,\infty})$ and $G_2 \in H^2(S_{-\infty,b})$.

(c) Show that $\lim_{a<y<b, y\to a} F(x+iy) = F_a(x)$ exists in the L^2-norm and also almost everywhere, with a similar result for $\lim_{a<y<b, y\to b} F(x+iy)$.

6. Suppose Ω is an open set in $\mathbb{C} = \mathbb{R}^2$, and let $\mathcal{H}$ be the subspace of $L^2(\Omega)$ consisting of holomorphic functions on Ω. Show that $\mathcal{H}$ is a closed subspace of $L^2(\Omega)$, and hence is a Hilbert space with inner product

$$(f, g) = \int_\Omega f(z)\overline{g}(z)\, dx\, dy, \qquad \text{where } z = x + iy.$$

[Hint: Prove that for $f \in \mathcal{H}$, we have $|f(z)| \leq \frac{c}{d(z,\Omega^c)}\|f\|$ for $z \in \Omega$, where $c = \pi^{-1/2}$, using the mean-value property (9). Thus if $\{f_n\}$ is a Cauchy sequence in $\mathcal{H}$, it converges uniformly on compact subsets of Ω.]

7. Following up on the previous exercise, prove:

(a) If $\{\varphi_n\}_{n=0}^{\infty}$ is an orthonormal basis of $\mathcal{H}$, then

$$\sum_{n=0}^{\infty} |\varphi_n(z)|^2 \leq \frac{c^2}{d(z, \Omega^c)} \qquad \text{for } z \in \Omega.$$

(b) The sum

$$B(z, w) = \sum_{n=0}^{\infty} \varphi_n(z)\overline{\varphi_n}(w)$$

converges absolutely for $(z, w) \in \Omega \times \Omega$, and is independent of the choice of the orthonormal basis $\{\varphi_n\}$ of $\mathcal{H}$.

(c) To prove (b) it is useful to characterize the function $B(z, w)$, called the **Bergman kernel**, by the following property. Let T be the linear transformation on $L^2(\Omega)$ defined by

$$Tf(z) = \int_{\Omega} B(z, w) f(w)\, du\, dv, \qquad w = u + iv.$$

Then T is the orthogonal projection of $L^2(\Omega)$ to $\mathcal{H}$.

(d) Suppose that Ω is the unit disc. Then $f \in \mathcal{H}$ exactly when $f(z) = \sum_{n=0}^{\infty} a_n z^n$, with

$$\sum_{n=0}^{\infty} |a_n|^2 (n+1)^{-1} < \infty.$$

Also, the sequence $\{\frac{z^n(n+1)}{\pi^{1/2}}\}_{n=0}^{\infty}$ is an orthonormal basis of $\mathcal{H}$. Moreover, in this case

$$B(z, w) = \frac{1}{\pi(1 - z\overline{w})^2}.$$

8. Continuing with Exercise 6, suppose Ω is the upper half-plane $\mathbb{R}^2_+$. Then every $f \in \mathcal{H}$ has a representation

$$f(z) = \sqrt{4\pi} \int_0^\infty \hat{f}_0(\xi) e^{2\pi i \xi z} \, d\xi, \qquad z \in \mathbb{R}^2_+, \tag{34}$$

where $\int_0^\infty |\hat{f}_0(\xi)|^2 \frac{d\xi}{\xi} < \infty$. Moreover, the mapping $\hat{f}_0 \to f$ given by (34) is a unitary mapping from $L^2((0,\infty), \frac{d\xi}{\xi})$ to $\mathcal{H}$.

9. Let H be the Hilbert transform. Verify that

(a) $H^* = -H$, $H^2 = -I$, and H is unitary.

(b) If τ_h denotes the translation operator, $\tau_h(f)(x) = f(x-h)$, then H commutes with τ_h, $\tau_h H = H\tau_h$.

(c) If δ_a denotes the dilation operator, $\delta_a(f)(x) = f(ax)$ with $a > 0$, then H commutes with δ_a, $\delta_a H = H\delta_a$.

A converse is given in Problem 5 below.

10. Let $f \in L^2(\mathbb{R})$ and let $u(x,y)$ be the Poisson integral of f, that is $u = (f * \mathcal{P}_y)(x)$, as given in (10) above. Let $v(x,y) = (Hf * \mathcal{P}_y)(x)$, the Poisson integral of the Hilbert transform of f. Prove that:

(a) $F(x+iy) = u(x,y) + iv(x,y)$ is analytic in the half-plane $\mathbb{R}^2_+$, so that u and v are conjugate harmonic functions. We also have $f = \lim_{y\to 0} u(x,y)$ and $Hf = \lim_{y\to 0} v(x,y)$.

(b) $F(z) = \frac{1}{\pi i}\int_{\mathbb{R}} f(t)\frac{dt}{t-z}$.

(c) $v(x,y) = f * \mathcal{Q}_y$, where $\mathcal{Q}_y(x) = \frac{1}{\pi}\frac{x}{x^2+y^2}$ is the **conjugate Poisson kernel**.

[Hint: Note that $\frac{i}{\pi z} = \mathcal{P}_y(x) + i\mathcal{Q}_y(x)$, $z = x+iy$.]

11. Show that

$$\left\{ \frac{1}{\pi^{1/2}(i+z)} \left(\frac{i-z}{i+z} \right)^n \right\}_{n=0}^{\infty}$$

is an orthonormal basis of $H^2(\mathbb{R}^2_+)$.

Note that $\left\{ \frac{1}{\pi^{1/2}(i+x)} \left(\frac{i-x}{i+x} \right)^n \right\}_{n=0}^{\infty}$ is an orthonormal basis of $L^2(\mathbb{R})$; see Exercise 9 in the previous chapter.

[Hint: It suffices to show that if $F \in H^2(\mathbb{R}^2_+)$ and

$$\int_{-\infty}^{\infty} F(x) \frac{(x+i)^n}{(x-i)^{n+1}} \, dx = 0 \qquad \text{for } n = 0, 1, 2, \ldots,$$

then $F = 0$. Use the Cauchy integral formula to prove that

$$\left(\frac{d}{dz} \right)^n (F(z)(z+i)^n)|_{z=i} = 0,$$

and thus $F^{(n)}(i) = 0$ for $n = 0, 1, 2, \ldots$.]

12. We consider whether the inequality

$$\|u\|_{L^2(\Omega)} \leq c\|L(u)\|_{L^2(\Omega)}$$

can hold for open sets Ω that are unbounded.

(a) Assume $d \geq 2$. Show that for each constant coefficient partial differential operator L, there are unbounded connected open sets Ω for which the above holds for all $u \in C_0^\infty(\Omega)$.

(b) Show that $\|u\|_{L^2(\mathbb{R}^d)} \leq c\|L(u)\|_{L^2(\mathbb{R}^d)}$ for all $u \in C_0^\infty(\mathbb{R}^d)$ if and only if $|P(\xi)| \geq c > 0$ all ξ, where P is the characteristic polynomial of L.

[Hint: For (a) consider first $L = (\partial/\partial x_1)^n$ and a strip $\{x : \ -1 < x_1 < 1\}$.]

13. Suppose L is a linear partial differential operator with constant coefficients. Show that when $d \geq 2$, the linear space of solutions u of $L(u) = 0$ with $u \in C^\infty(\mathbb{R}^d)$ is not finite-dimensional.

[Hint: Consider the zeroes ζ of $P(\zeta)$, $\zeta \in \mathbb{C}^d$, where P is the characteristic polynomial of L.]

14. Suppose F and G are two integrable functions on a bounded interval $[a, b]$. Show that G is the weak derivative of F if and only if F can be corrected on a set of measure 0, such that F is absolutely continuous and $F'(x) = G(x)$ for almost every x.

[Hint: If G is the weak derivative of F, use an approximation to show that

$$\int_a^b G(x)\varphi(x)dx = -\int_a^b F(x)\varphi'(x)dx$$

holds for the function φ illustrated in Figure 9.]

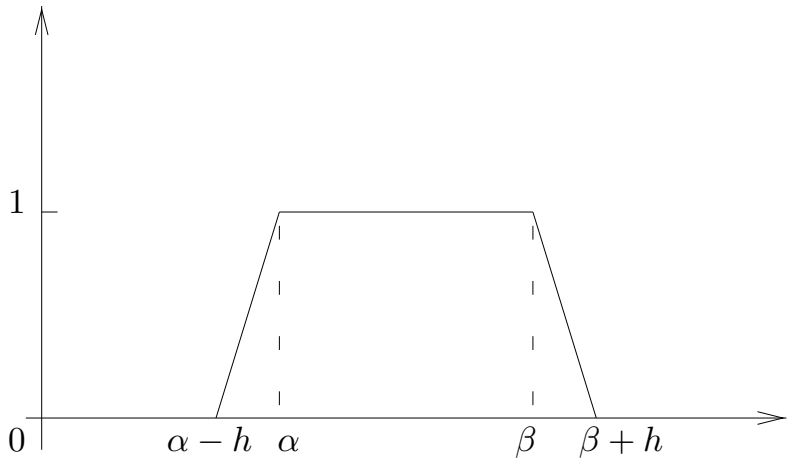


Figure 9. The function φ in Exercise 14

15. Suppose $f \in L^2(\mathbb{R}^d)$. Prove that there exists $g \in L^2(\mathbb{R}^d)$ such that

$$\left(\frac{\partial}{\partial x}\right)^{\alpha} f(x) = g(x)$$

in the weak sense, if and only if

$$(2\pi i\xi)^{\alpha} \hat{f}(\xi) = \hat{g}(\xi) \in L^2(\mathbb{R}^d).$$

16. Sobolev embedding theorem. Suppose n is the smallest integer $> d/2$. If

$$f \in L^2(\mathbb{R}^d) \quad \text{and} \quad \left(\frac{\partial}{\partial x}\right)^\alpha f \in L^2(\mathbb{R}^d)$$

in the weak sense, for all $1 \leq |\alpha| \leq n$, then f can be modified on a set of measure zero so that f is continuous and bounded.

[Hint: Express f in terms of $\hat{f}$, and show that $\hat{f} \in L^1(\mathbb{R}^d)$ by the Cauchy-Schwarz inequality.]

17. The conclusion of the Sobolev embedding theorem fails when $n = d/2$. Consider the case $d = 2$, and let $f(x) = (\log 1/|x|)^\alpha \eta(x)$, where η is a smooth cut-off function with $\eta = 1$ for x near the origin, but $\eta(x) = 0$ if $|x| \geq 1/2$. Let $0 < \alpha < 1/2$.

(a) Verify that $\partial f/\partial x_1$ and $\partial f/\partial x_2$ are in L^2 in the weak sense.

(b) Show that f cannot be corrected on a set of measure zero such that the resulting function is continuous at the origin.

18. Consider the linear partial differential operator

$$L = \sum_{|\alpha| \le n} a_\alpha \left(\frac{\partial}{\partial x} \right)^\alpha .$$

Then

$$P(\xi) = \sum_{|\alpha| \le n} a_\alpha (2\pi i \xi)^\alpha$$

is called the **characteristic polynomial** of L. The differential operator L is said to be **elliptic** if

$$|P(\xi)| \ge c|\xi|^n \qquad \text{for some } c > 0 \text{ and all } \xi \text{ sufficiently large.}$$

(a) Check that L is elliptic if and only if $\sum_{|\alpha|=n} a_\alpha (2\pi\xi)^\alpha$ vanishes only when $\xi = 0$.

(b) If L is elliptic, prove that for some $c > 0$ the inequality

$$\left\| \left(\frac{\partial}{\partial x} \right)^\alpha \varphi \right\|_{L^2(\mathbb{R}^d)} \le c \left(\|L\varphi\|_{L^2(\mathbb{R}^d)} + \|\varphi\|_{L^2(\mathbb{R}^d)} \right)$$

holds for all $\varphi \in C_0^\infty(\Omega)$ and $|\alpha| \le n$.

(c) Conversely, if (b) holds then L is elliptic.

19. Suppose u is harmonic in the punctured unit disc $\mathbb{D}^* = \{z \in \mathbb{C} : 0 < |z| < 1\}$.

(a) Show that if u is also continuous at the origin, then u is harmonic throughout the unit disc.

[Hint: Show that u is weakly harmonic.]

(b) Prove that the Dirichlet problem for the punctured unit disc is in general not solvable.

20. Let F be a continuous function on the closure $\overline{\mathbb{D}}$ of the unit disc. Assume that F is in C^1 on the (open) disc $\mathbb{D}$, and $\int_{\mathbb{D}} |\nabla F|^2 < \infty$.

Let $f(e^{i\theta})$ denote the restriction of F to the unit circle, and write $f(e^{i\theta}) \sim \sum_{n=-\infty}^{\infty} a_n e^{in\theta}$. Prove that $\sum_{n=-\infty}^{\infty} |n|\,|a_n|^2 < \infty$.

[Hint: Write $F(re^{i\theta}) \sim \sum_{n=-\infty}^{\infty} F_n(r) e^{in\theta}$, with $F_n(1) = a_n$. Express $\int_{\mathbb{D}} |\nabla F|^2$ in polar coordinates, and use the fact that

$$\frac{1}{2}|F(1)|^2 \le L^{-1} \int_{1/2}^{1} |F'(r)|^2\, dr + L \int_{1/2}^{1} |F(r)|^2\, dr,$$

for $L \ge 2$; apply this to $F = F_n$, $L = |n|$.]

Chapter 6. Abstract Measure and Integration Theory

1. Let X be a set and $\mathcal{M}$ a non-empty collection of subsets of X. Prove that if $\mathcal{M}$ is closed under complements and countable unions of disjoint sets, then $\mathcal{M}$ is a σ-algebra.

[Hint: Any countable union of sets can be written as a countable union of disjoint sets.]

2. Let $(X, \mathcal{M}, \mu)$ be a measure space. One can define the **completion** of this space as follows. Let $\overline{\mathcal{M}}$ be the collection of sets of the form $E \cup Z$, where $E \in \mathcal{M}$, and $Z \subset F$ with $F \in \mathcal{M}$ and $\mu(F) = 0$. Also, define $\overline{\mu}(E \cup Z) = \mu(E)$. Then:

(a) $\overline{\mathcal{M}}$ is the smallest σ-algebra containing $\mathcal{M}$ and all subsets of elements of $\mathcal{M}$ of measure zero.

(b) The function $\overline{\mu}$ is a measure on $\overline{\mathcal{M}}$, and this measure is complete.

[Hint: To prove $\overline{\mathcal{M}}$ is a σ-algebra it suffices to see that if $E_1 \subset \overline{\mathcal{M}}$, then $E_1^c \subset \overline{\mathcal{M}}$. Write $E_1 = E \cup Z$ with $Z \subset F$, E and F in $\mathcal{M}$. Then $E_1^c = (E \cup F)^c \cup (F - Z)$.]

3. Consider the exterior Lebesgue measure m_* introduced in Chapter 1. Prove that a set E in $\mathbb{R}^d$ is Carathéodory measurable if and only if E is Lebesgue measurable in the sense of Chapter 1.

[Hint: If E is Lebesgue measurable and A is any set, choose a G_δ set G such that $A \subset G$ and $m_*(A) = m(G)$. Conversely, if E is Carathéodory measurable and $m_*(E) < \infty$, choose a G_δ set G with $E \subset G$ and $m_*(E) = m_*(G)$. Then $G - E$ has exterior measure 0.]

4. Let r be a rotation of $\mathbb{R}^d$. Using the fact that the mapping $x \mapsto r(x)$ preserves Lebesgue measure (see Problem 4 in Chapter 2 and Exercise 26 in Chapter 3), show that it induces a measure-preserving map of the sphere S^{d-1} with its measure $d\sigma$.

A converse is stated in Problem 4.

5. Use the polar coordinate formula to prove the following:

(a) $\int_{\mathbb{R}^d} e^{-\pi|x|^2}\,dx = 1$, when $d = 2$. Deduce from this that the same identity holds for all d.

(b) $\left(\int_0^\infty e^{-\pi r^2} r^{d-1}\,dr\right)\sigma(S^{d-1}) = 1$, and as a result, $\sigma(S^{d-1}) = 2\pi^{d/2}/\Gamma(d/2)$.

(c) If B is the unit ball, $v_d = m(B) = \pi^{d/2}/\Gamma(d/2+1)$, since this quantity equals $\left(\int_0^1 r^{d-1}\,dr\right)\sigma(S^{d-1})$. (See Exercise 14 in Chapter 2.)

6. A version of Green's formula for the unit ball B in $\mathbb{R}^d$ can be stated as follows. Suppose u and v are a pair of functions that are in $C^2(\overline{B})$. Then one has

$$\int_B (v\triangle u - u\triangle v)\, dx = \int_{S^{d-1}} \left(v\frac{\partial u}{\partial n} - u\frac{\partial v}{\partial n} \right) d\sigma.$$

Here S^{d-1} is the unit sphere with $d\sigma$ the measure defined in Section 3.2, and $\partial u/\partial n$, $\partial v/\partial n$ denote the directional derivatives of u and v (respectively) along the inner normals to S^{d-1}.

Show that the above can be derived from Lemma 4.5 of the previous chapter by taking $\eta = \eta_\epsilon^+$ and letting $\epsilon \to 0$.

7. There is an alternate version of the mean-value property given in (21) of Chapter 5. It can be stated as follows. Suppose u is harmonic in Ω, and B is any ball of center x_0 and radius r whose closure is contained in Ω. Then

$$u(x_0) = c \int_{S^{d-1}} u(x_0 + ry)\, d\sigma(y), \qquad \text{with } c^{-1} = \sigma(S^{d-1}).$$

Conversely, a continuous function satisfying this mean-value property is harmonic.

[Hint: This can be proved as a direct consequence of the corresponding result for averages over balls (Theorem 4.27 in Chapter 5), or can be deduced from Exercise 6.]

8. The fact that the Lebesgue measure is uniquely characterized by its translation invariance can be made precise by the following assertion: If μ is a Borel measure on $\mathbb{R}^d$ that is translation-invariant, and is finite on compact sets, then μ is a multiple of Lebesgue measure m. Prove this theorem by proceeding as follows.

(a) Suppose Q_a denotes a translate of the cube $\{x : 0 < x_j \leq a,\ j = 1, 2, \ldots, d\}$ of side length a. If we let $\mu(Q_1) = c$, then $\mu(Q_{1/n}) = cn^{-d}$ for each integer n.

(b) As a result μ is absolutely continuous with respect to m, and there is a locally integrable function f such that

$$\mu(E) = \int_E f\, dx.$$

(c) By the differentiation theorem (Corollary 1.7 in Chapter 3) it follows that $f(x) = c$ a.e., and hence $\mu = cm$.

[Hint: Q_1 can be written as a disjoint union of n^d translates of $Q_{1/n}$.]

9. Let $C([a,b])$ denote the vector space of continuous functions on the closed and bounded interval $[a,b]$. Suppose we are given a Borel measure μ on this interval, with $\mu([a,b]) < \infty$. Then

$$f \mapsto \ell(f) = \int_a^b f(x)\, d\mu(x)$$

is a linear functional on $C([a,b])$, with ℓ positive in the sense that $\ell(f) \geq 0$ if $f \geq 0$.

Prove that, conversely, for any linear functional ℓ on $C([a,b])$ that is positive in the above sense, there is a unique finite Borel measure μ so that $\ell(f) = \int_a^b f\, d\mu$ for $f \in C([a,b])$.

[Hint: Suppose $a = 0$ and $u \geq 0$. Define $F(u)$ by $F(u) = \lim_{\epsilon \to 0} \ell(f_\epsilon)$, where

$$f_\epsilon(x) = \begin{cases} 1 & \text{for } 0 \leq x \leq u, \\ 0 & \text{for } u + \epsilon \leq x, \end{cases}$$

and f_ϵ is linear between u and $u + \epsilon$. (See Figure 3.) Then F is increasing and right-continuous, and $\ell(f)$ can be written as $\int_a^b f(x)\, dF(x)$ via Theorem 3.5.]

The result also holds if $[a,b]$ is replaced by a closed infinite interval; we then assume that ℓ is defined on the continuous functions of bounded support, and obtain that the resulting μ is finite on all bounded intervals.

A generalization is given in Problem 5.

10. Suppose ν, ν_1, ν_2 are signed measures on $(X, \mathcal{M})$ and μ a (positive) measure on $\mathcal{M}$. Using the symbols $\perp$ and $\ll$ defined in Section 4.2, prove:

(a) If $\nu_1 \perp \mu$ and $\nu_2 \perp \mu$, then $\nu_1 + \nu_2 \perp \mu$.

(b) If $\nu_1 \ll \mu$ and $\nu_2 \ll \mu$, then $\nu_1 + \nu_2 \ll \mu$.

(c) $\nu_1 \perp \nu_2$ implies $|\nu_1| \perp |\nu_2|$.

(d) $\nu \ll |\nu|$.

(e) If $\nu \perp \mu$ and $\nu \ll \mu$, then $\nu = 0$.

11. Suppose that F is an increasing normalized function on $\mathbb{R}$, and let $F = F_A + F_C + F_J$ be the decomposition of F in Exercise 24 in Chapter 3; here F_A is

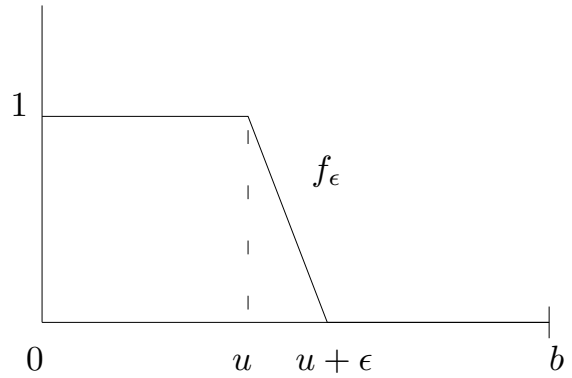


Figure 3. The function f_ϵ in Exercise 9

absolutely continuous, F_C is continuous with $F_C' = 0$ a.e, and F_J is a pure jump function. Let $\mu = \mu_A + \mu_C + \mu_J$ with μ, μ_A, μ_C, and μ_J the Borel measures associated to F, F_A, F_C, and F_J, respectively. Verify that:

(i) μ_A is absolutely continuous with respect to Lebesgue measure and $\mu_A(E) = \int_E F'(x)\,dx$ for every Lebesgue measurable set E.

(ii) As a result, if F is absolutely continuous, then $\int f\,d\mu = \int f\,dF = \int f(x)F'(x)\,dx$ whenever f and fF' are integrable.

(iii) $\mu_C + \mu_J$ and Lebesgue measure are mutually singular.

12. Suppose $\mathbb{R}^d - \{0\}$ is represented as $\mathbb{R}_+ \times S^{d-1}$, with $\mathbb{R}_+ = \{0 < r < \infty\}$. Then every open set in $\mathbb{R}^d - \{0\}$ can be written as a countable union of open rectangles of this product.

[Hint: Consider the countable collection of rectangles of the form

$$\{r_j < r < r'_k\} \times \{\gamma \in S^{d-1} : \ |\gamma - \gamma_\ell| < 1/n\}.$$

Here r_j and r'_k range over all positive rationals, and $\{\gamma_\ell\}$ is a countable dense set of S^{d-1}.]

13. Let m_j be the Lebesgue measure for the space $\mathbb{R}^{d_j}$, $j = 1, 2$. Consider the product $\mathbb{R}^d = \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$ $(d = d_1 + d_2)$, with m the Lebesgue measure on $\mathbb{R}^d$. Show that m is the completion (in the sense of Exercise 2) of the product measure $m_1 \times m_2$.

14. Suppose $(X_j, \mathcal{M}_j, \mu_j)$, $1 \le j \le k$, is a finite collection of measure spaces. Show that parallel with the case $k = 2$ considered in Section 3 one can construct a product measure $\mu_1 \times \mu_2 \times \cdots \times \mu_k$ on $X = X_1 \times X_2 \times \cdots \times X_k$. In fact, for any set $E \subset X$ such that $E = E_1 \times E_2 \times \cdots \times E_k$, with $E_j \subset \mathcal{M}_j$ for all j, define $\mu_0(E) = \prod_{j=1}^{k} \mu_j(E_j)$. Verify that μ_0 extends to a premeasure on the algebra $\mathcal{A}$ of finite disjoint unions of such sets, and then apply Theorem 1.5.

15. The product theory extends to infinitely many factors, under the requisite assumptions. We consider measure spaces $(X_j, \mathcal{M}_j, \mu_j)$ with $\mu_j(X_j) = 1$ for all but finitely many j. Define a **cylinder set** E as

$$\{x = (x_j),\ x_j \in E_j,\ E_j \in \mathcal{M}_j, \text{but } E_j = X_j \text{ for all but finitely many } j\}.$$

For such a set define $\mu_0(E) = \prod_{j=1}^{\infty} \mu_j(E_j)$. If $\mathcal{A}$ is the algebra generated by the cylinder sets, μ_0 extends to a premeasure on $\mathcal{A}$, and we can apply Theorem 1.5 again.

16. Consider the d-dimensional torus $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$. Identify $\mathbb{T}^d$ as $\mathbb{T}^1 \times \cdots \times \mathbb{T}^1$ (d factors) and let μ be the product measure on $\mathbb{T}^d$ given by $\mu = \mu_1 \times \mu_2 \times \cdots \times \mu_d$, where μ_j is Lebesgue measure on X_j identified with the circle $\mathbb{T}$. That is, if we represent each point in X_j uniquely as x_j with $0 < x_j \leq 1$, then the measure μ_j is the induced Lebesgue measure on $\mathbb{R}^1$ restricted to $(0, 1]$.

(a) Check that the completion μ is Lebesgue measure induced on the cube $Q = \{x : \ 0 < x_j \leq 1, \ j = 1, \ldots, d\}$.

(b) For each function f on Q let $\tilde{f}$ be its extension to $\mathbb{R}^d$ which is periodic, that is, $\tilde{f}(x + z) = \tilde{f}(x)$ for every $z \in \mathbb{Z}^d$. Then f is measurable on $\mathbb{T}^d$ if and only if $\tilde{f}$ is measurable on $\mathbb{R}^d$, and f is continuous on $\mathbb{T}^d$ if and only if $\tilde{f}$ is continuous on $\mathbb{R}^d$.

(c) Suppose f and g are integrable on $\mathbb{T}^d$. Show that the integral defining $(f * g)(x) = \int_{\mathbb{T}^d} f(x - y)g(y)\, dy$ is finite for a.e. x, that $f * g$ is integrable over $\mathbb{T}^d$, and that $f * g = g * f$.

(d) For any integrable function f on $\mathbb{T}^d$, write

$$f \sim \sum_{n \in \mathbb{Z}^d} a_n e^{2\pi i n \cdot x}$$

to mean that $a_n = \int_{\mathbb{T}^d} f(x) e^{-2\pi i n \cdot x}\, dx$. Prove that if g is also integrable, and $g \sim \sum_{n \in \mathbb{Z}^d} b_n e^{2\pi i n \cdot x}$, then

$$f * g \sim \sum_{n \in \mathbb{Z}^d} a_n b_n e^{2\pi i n \cdot x}.$$

(e) Verify that $\{e^{2\pi i n \cdot x}\}_{n \in \mathbb{Z}^d}$ is an orthonormal basis for $L^2(\mathbb{T}^d)$. As a result $\|f\|_{L^2(\mathbb{T}^d)} = \sum_{n \in \mathbb{Z}^d} |a_n|^2$.

(f) Let f be any continuous periodic function on $\mathbb{T}^d$. Then f can be uniformly approximated by finite linear combinations of the exponentials $\{e^{2\pi i n \cdot x}\}_{n \in \mathbb{Z}^d}$.

[Hint: For (e), reduce to the case $d = 1$ by Fubini's theorem. To prove (f) let $g(x) = g_\epsilon(x) = \epsilon^{-d}$, if $0 < x_j \leq \epsilon$, $j = 1, \ldots, d$, and $g_\epsilon(x) = 0$ elsewhere in Q. Then $(f * g_\epsilon)(x) \to f(x)$ uniformly as $\epsilon \to 0$. However $(f * g_\epsilon)(x) = \sum a_n b_n e^{2\pi i n x}$ with $b_n = \int_{\mathbb{T}^d} g_\epsilon(x) e^{-2\pi i n \cdot x}\, dx$, and $\sum |a_n b_n| < \infty$.]

17. By reducing to the case $d = 1$, show that each "rotation" $x \mapsto x + \alpha$ of the torus $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$ is measure preserving, for any $\alpha \in \mathbb{R}^d$.

18. Suppose τ is a measure-preserving transformation on a measure space (X, μ) with $\mu(X) = 1$. Recall that a measurable set E is invariant if $\tau^{-1}(E)$ and E differ by a set of measure zero. A sharper notion is to require that $\tau^{-1}(E)$ equal E. Prove that if E is any invariant set, there is a set E' so that $E' = \tau^{-1}(E')$, and E and E' differ by a set of measure zero.

[Hint: Let $E' = \limsup_{n\to\infty}\{\tau^{-n}(E)\} = \bigcap_{n=0}^{\infty}\left(\bigcup_{k\geq n}\tau^{-k}(E)\right)$.]

19. Let τ be a measure-preserving transformation on (X, μ) with $\mu(X) = 1$. Then τ is ergodic if and only if whenever ν is absolutely continuous with respect to μ and ν is invariant (that is, $\nu(\tau^{-1}(E)) = \nu(E)$ for all measurable sets E), then $\nu = c\mu$, with c a constant.

20. Suppose τ is a measure-preserving transformation on (X, μ). If

$$\mu(\tau^{-n}(E) \cap F) \to \mu(E)\mu(F)$$

as $n \to \infty$ for all measurable sets E and F, then $(T^n f, g) \to (f, 1)(1, g)$ whenever $f, g \in L^2(X)$ with $(Tf)(x) = f(\tau(x))$. Thus τ is mixing.

[Hint: By linearity the hypothesis implies the conclusion whenever f and g are simple functions.]

21. Let $\mathbb{T}^d$ be the torus, and $\tau : x \mapsto x + \alpha$ the mapping arising in Exercise 17. Then τ is ergodic if and only if $\alpha = (\alpha_1, \ldots, \alpha_d)$ with $\alpha_1, \alpha_2, \ldots, \alpha_d$, and 1 are linearly independent over the rationals. To do this show that:

(a) $\dfrac{1}{m} \displaystyle\sum_{k=0}^{m-1} f(\tau^k(x)) \to \int_{\mathbb{T}^d} f(x)\, dx$ as $m \to \infty$, for each $x \in \mathbb{T}^d$, whenever f is continuous and periodic and α satisfies the hypothesis.

(b) Prove as a result that in this case τ is uniquely ergodic.

[Hint: Use (f) in Exercise 16.]

22. Let $X = \prod_{i=1}^{\infty} X_i$, where each (X_i, μ_i) is identical to (X_1, μ_1), with $\mu_1(X_1) = 1$, and let μ be the corresponding product measure defined in Exercise 15. Define the **shift** $\tau : X \to X$ by $\tau((x_1, x_2, \ldots)) = (x_2, x_3, \ldots)$ for $x = (x_i) \in \prod_{i=1}^{\infty} X_i$.

(a) Verify that τ is a measure-preserving transformation.

(b) Prove that τ is ergodic by showing that it is mixing.

(c) Note that in general τ is not uniquely ergodic.

If we define the corresponding shift on the two-sided infinite product, then τ is also a measure-preserving isomorphism.

[Hint: For (b) note that $\mu(\tau^{-n}(E \cap F)) = \mu(E)\mu(F)$ whenever E and F are cylinder sets and n is sufficiently large. For (c) note that, for example, if we fix a point $\overline{x} \in X_1$, the set $E = \{(x_i) : \ x_j = \overline{x} \text{ all } j\}$ is invariant.]

23. Let $X = \prod_{i=1}^{\infty} Z(2)$, where each factor is the two-point space $Z(2) = \{0, 1\}$ with $\mu_1(0) = \mu_1(1) = 1/2$, and suppose μ denotes the product measure on X. Consider the mapping $D : X \to [0, 1]$ given by $D(\{a_j\}) \to \sum_{j=1}^{\infty} \frac{a_j}{2^j}$. Then there are denumerable sets $Z_1 \subset X$ and $Z_2 \subset [0, 1]$, such that:

(a) D is a bijection from $X - Z_1$ to $[0, 1] - Z_2$.

(b) A set E in X is measurable if and only if $D(E)$ is measurable in $[0, 1]$, and $\mu(E) = m(D(E))$, where m is Lebesgue measure on $[0, 1]$.

(c) The shift map on $\prod_{i=1}^{\infty} Z(2)$ then becomes the doubling map of example (b) in Section 5.4.

24. Consider the following generalization of the doubling map. For each integer m, $m \geq 2$, we define the map τ_m of $(0, 1]$ by $\tau(x) = mx \mod 1$.

(a) Verify that τ is measure-preserving for Lebesgue measure.

(b) Show that τ is mixing, hence ergodic.

(c) Prove as a consequence that almost every number x is normal in the scale m, in the following sense. Consider the m-adic expansion of x,

$$x = \sum_{j=1}^{\infty} \frac{a_j}{m^j}, \qquad \text{where each } a_j \text{ is an integer } 0 \leq a_j \leq m-1.$$

Then x is **normal** if for each integer k, $0 \leq k \leq m-1$,

$$\frac{\#\{j : \ a_j = k, \ 1 \leq j \leq n\}}{N} \to \frac{1}{m} \qquad \text{as } N \to \infty.$$

Note the analogy with the equidistribution statements in Section 2, Chapter 4, of Book I.

25. Show that the mean ergodic theorem still holds if we replace the assumption that T is an isometry by the assumption that T is a **contraction**, that is, $\|Tf\| \leq \|f\|$ for all $f \in \mathcal{H}$.

[Hint: Prove that T is a contraction if and only if T^* is a contraction, and use the identity $(f, T^*f) = (Tf, f)$.]

26. There is an L^2 version of the maximal ergodic theorem. Suppose τ is a measure-preserving transformation on (X, μ). Here we do not assume that $\mu(X) < \infty$. Then

$$f^*(x) = \sup \frac{1}{m} \sum_{k=0}^{m-1} |f(\tau^k(x))|$$

satisfies

$$\|f^*\|_{L^2(X)} \leq c\|f\|_{L^2(X)}, \qquad \text{whenever } f \in L^2(X).$$

The proof is the same as outlined in Problem 6, Chapter 5 for the maximal function on $\mathbb{R}^d$. With this, extend the pointwise ergodic theorem to the case where $\mu(X) = \infty$, as follows:

(a) Show that $\lim_{m\to\infty} \frac{1}{m} \sum_{k=0}^{m-1} f(\tau^k(x))$ converges for a.e. x to $P(f)(x)$ for every $f \in L^2(X)$, because this holds for a dense subspace of $L^2(X)$.

(b) Prove that the conclusion holds for every $f \in L^1(X)$, because it holds for the dense subspace $L^1(X) \cap L^2(X)$.

27. We saw that if $\|f_n\|_{L^2} \leq 1$, then $\frac{f_n(x)}{n} \to 0$ as $n \to \infty$ for a.e. x. However, show that the analogue where one replaces the L^2-norm by the L^1-norm fails, by constructing a sequence $\{f_n\}$, $f_n \in L^1(X)$, $\|f_n\|_{L^1} \leq 1$, but with $\limsup_{n\to\infty} \frac{f_n(x)}{n} = \infty$ for a.e. x.

[Hint: Find intervals $I_n \subset [0,1]$, so that $m(I_n) = 1/(n \log n)$ but $\limsup_{n\to\infty}\{I_n\} = [0,1]$. Then take $f_n(x) = n \log n \chi_{I_n}$.]

28. We know by the Borel-Cantelli lemma that if $\{E_n\}$ is a collection of measurable sets in a measure a space (X, μ) and $\sum_{n=1}^{\infty} \mu(E_n) < \infty$ then $E = \limsup_{n\to\infty}\{E_n\}$ has measure zero.

In the opposite direction, if τ is a mixing measure-preserving transformation on X (with $\mu(X) = 1$), then whenever $\sum_{n=1}^{\infty} \mu(E_n) = \infty$, there are integers $m = m_n$ so that if $E'_n = \tau^{-m_n}(E_n)$, then $\limsup_{n\to\infty}(E'_n) = X$, except for a set of measure 0.

Chapter 7. Hausdorff Measure and Fractals

1. Show that the measure m_α is not σ-finite on $\mathbb{R}^d$ if $\alpha < d$.

2. Suppose E_1 and E_2 are two compact subsets of $\mathbb{R}^d$ such that $E_1 \cap E_2$ contains at most one point. Show directly from the definition of the exterior measure that if $0 < \alpha \leq d$, and $E = E_1 \cup E_2$, then

$$m_\alpha^*(E) = m_\alpha^*(E_1) + m_\alpha^*(E_2).$$

[Hint: Suppose $E_1 \cap E_2 = \{x\}$, let B_ϵ denote the open ball centered at x and of diameter ϵ, and let $E^\epsilon = E \cap B_\epsilon^c$. Show that

$$m_\alpha^*(E^\epsilon) \geq \mathcal{H}_\alpha^\epsilon(E) \geq m_\alpha^*(E) - \mu(\epsilon) - \epsilon^\alpha,$$

where $\mu(\epsilon) \to 0$. Hence $m_\alpha^*(E^\epsilon) \to m_\alpha^*(E)$.]

3. Prove that if $f : [0, 1] \to \mathbb{R}$ satisfies a Lipschitz condition of exponent $\gamma > 1$, then f is a constant.

4. Suppose $f : [0,1] \to [0,1] \times [0,1]$ is surjective and satisfies a Lipschitz condition

$$|f(x) - f(y)| \leq C|x - y|^{\gamma}.$$

Prove that $\gamma \leq 1/2$ directly, without using Theorem 2.2.

[Hint: Divide $[0,1]$ into N intervals of equal length. The image of each sub-interval is contained in a ball of volume $O(N^{-2\gamma})$, and the union of all these balls must cover the square.]

5. Let $f(x) = x^k$ be defined on $\mathbb{R}$, where k is a positive integer and let E be a Borel subset of $\mathbb{R}$.

(a) Show that if $m_\alpha(E) = 0$ for some α, then $m_\alpha(f(E)) = 0$.

(b) Prove that $\dim(E) = \dim f(E)$.

6. Let $\{E_k\}$ be a sequence of Borel sets in $\mathbb{R}^d$. Show that if $\dim E_k \le \alpha$ for some α and all k, then

$$\dim \bigcup_k E_k \le \alpha.$$

7. Prove that the $(\log 2/\log 3)$-Hausdorff measure of the Cantor set is precisely equal to 1.

[Hint: Suppose we have a covering of $\mathcal{C}$ by finitely many closed intervals $\{I_j\}$. Then there exists another covering of $\mathcal{C}$ by intervals $\{I'_\ell\}$ each of length 3^{-k} for some k, such that $\sum_j |I_j|^\alpha \geq \sum_\ell |I'_\ell|^\alpha \geq 1$, where $\alpha = \log 2/\log 3$.]

8. Show that the Cantor set of constant dissection, $\mathcal{C}_\xi$, in Exercise 3 of Chapter 1 has strict Hausdorff dimension $\log 2/\log(2/(1-\xi))$.

9. Consider the set $\mathcal{C}_{\xi_1} \times \mathcal{C}_{\xi_2}$ in $\mathbb{R}^2$, with $\mathcal{C}_\xi$ as in the previous exercise. Show that $\mathcal{C}_{\xi_1} \times \mathcal{C}_{\xi_2}$ has strict Hausdorff dimension $\dim(\mathcal{C}_{\xi_1}) + \dim(\mathcal{C}_{\xi_2})$.

10. Construct a Cantor-like set (as in Exercise 4, Chapter 1) that has Lebesgue measure zero, yet Hausdorff dimension 1.

[Hint: Choose $\ell_1, \ell_2, \ldots, \ell_k, \ldots$ so that $1 - \sum_{j=1}^{k} 2^{j-1}\ell_j$ tends to zero sufficiently slowly as $k \to \infty$.]

11. Let $\mathcal{D} = \mathcal{D}_\mu$ be the Cantor dust in $\mathbb{R}^2$ given as the product $\mathcal{C}_\xi \times \mathcal{C}_\xi$, with $\mu = (1 - \xi)/2$.

(a) Show that for any real number λ, the set $\mathcal{C}_\xi + \lambda\mathcal{C}_\xi$ is similar to the projection of $\mathcal{D}$ on the line in $\mathbb{R}^2$ with slope $\lambda = \tan\theta$.

(b) Note that among the Cantor sets $\mathcal{C}_\xi$, the value $\xi = 1/2$ is critical in the construction of the Besicovitch set in Section 4.4. In fact, prove that with $\xi > 1/2$, then $\mathcal{C}_\xi + \lambda\mathcal{C}_\xi$ has Lebesgue measure zero for *every* λ. See also Problem 10 below.

[Hint: $m_\alpha(\mathcal{C}_\xi + \lambda\mathcal{C}_\xi) < \infty$ for $\alpha = \dim \mathcal{D}_\mu$.]

12. Define a primitive one-dimensional "measure" $\tilde{m}_1$ as

$$\tilde{m}_1 = \inf \sum_{k=1}^{\infty} \text{diam } F_k, \qquad E \subset \bigcup_{k=1}^{\infty} F_k.$$

This is akin to the one-dimensional exterior measure m_α^*, $\alpha = 1$, except that no restriction is placed on the size of the diameters F_k.

Suppose I_1 and I_2 are two *disjoint* unit segments in $\mathbb{R}^d$, $d \geq 2$, with $I_1 = I_2 + h$, and $|h| < \epsilon$. Then observe that $\tilde{m}_1(I_1) = \tilde{m}_1(I_2) = 1$, while $\tilde{m}_1(I_1 \cup I_2) \leq 1 + \epsilon$. Thus

$$\tilde{m}_1(I_1 \cup I_2) < \tilde{m}_1(I_1) + \tilde{m}_1(I_2) \qquad \text{when } \epsilon < 1;$$

hence $\tilde{m}_1$ fails to be additive.

13. Consider the von Koch curve $\mathcal{K}^\ell$, $1/4 < \ell < 1/2$, as defined in Section 2.1. Prove for it the analogue of Theorem 2.7: the function $t \mapsto \mathcal{K}^\ell(t)$ satisfies a Lipschitz condition of exponent $\gamma = \log(1/\ell)/\log 4$. Moreover, show that the set $\mathcal{K}^\ell$ has strict Hausdorff dimension $\alpha = 1/\gamma$.

[Hint: Show that if $\mathcal{O}$ is the shaded open triangle indicated in Figure 14, then $\mathcal{O} \supset S_0(\mathcal{O}) \cup S_1(\mathcal{O}) \cup S_2(\mathcal{O}) \cup S_3(\mathcal{O})$, where $S_0(x) = \ell x$, $S_1(x) = \rho_\theta(\ell x) + a$, $S_2(x) = \rho_\theta^{-1}(\ell x) + c$, and $S_3(x) = \ell x + b$, with ρ_θ the rotation of angle θ. Note that the sets $S_j(\mathcal{O})$ are disjoint.]

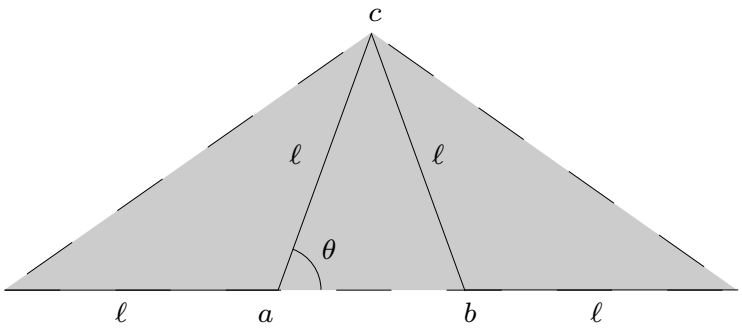


Figure 14. The open set $\mathcal{O}$ in Exercise 13

14. Show that if $\ell < 1/2$, the von Koch curve $t \mapsto \mathcal{K}^\ell(t)$ in Exercise 13 is a simple curve.

[Hint: Observe that if $t = \sum_{j=1}^{\infty} a_j/4^j$, with $a_j = 0, 1, 2$, or 3, then

$$\{\mathcal{K}(t)\} = \bigcap_{j=1}^{\infty} S_{a_j}\left(\cdots S_{a_2}\left(S_{a_1}(\overline{\mathcal{O}})\right)\right).]$$

15. Note that if we take $\ell = 1/2$ in the definition of the von Koch curve in Exercise 13 we get a "space-filling" curve, one that fills the right triangle whose vertices are $(0,0)$, $(1,0)$, and $(1/2,1/2)$. The first three steps of the construction are as in Figure 15, with the intervals traced out in the indicated order.

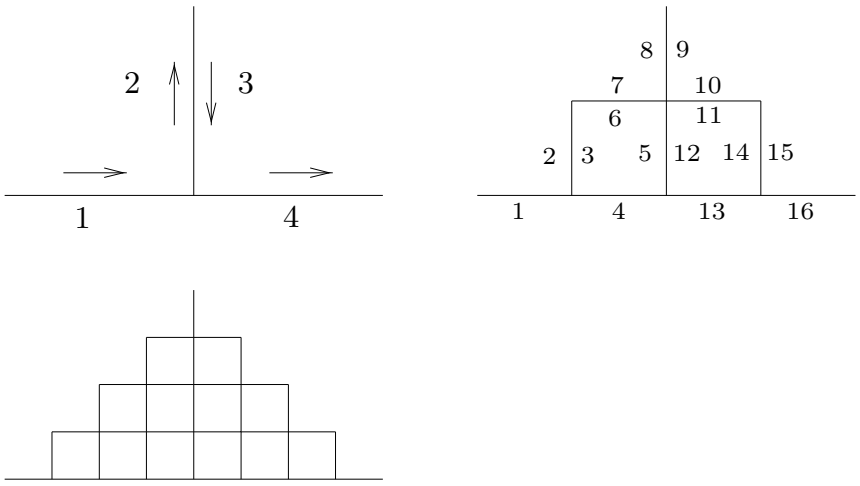


Figure 15. The first three steps of the von Koch curve when $\ell = 1/2$

16. Prove that the von Koch curve $t \mapsto \mathcal{K}^{\ell}(t)$, $1/4 < \ell \leq 1/2$ is continuous but nowhere differentiable.

[Hint: If $\mathcal{K}'(t)$ exists for some t, then

$$\lim_{n\to\infty} \frac{\mathcal{K}(u_n) - \mathcal{K}(v_n)}{u_n - v_n}$$

must exist, where $u_n \leq t \leq v_n$, and $u_n - v_n \to 0$. Choose $u_n = k/4^n$ and $v_n = (k+1)/4^n$.]

17. For a compact set E in $\mathbb{R}^d$, define $\#(\epsilon)$ to be the least number of balls of radius ϵ that cover E. Note that we always have $\#(\epsilon) = O(\epsilon^{-d})$ as $\epsilon \to 0$, and $\#(\epsilon) = O(1)$ if E is finite.

One defines the **covering dimension** of E, denoted by $\dim_C(E)$, as $\inf \beta$ such that $\#(\epsilon) = O(\epsilon^{-\beta})$, as $\epsilon \to 0$. Show that $\dim_C(E) = \dim_M(E)$, where $\dim_M$ is the Minkowski dimension discussed in Section 2.1, by proving the following inequalities for all $\delta > 0$:

(i) $m(E^\delta) \leq c\#(\delta)\delta^d$.

(ii) $\#(\delta) \leq c' m(E^\delta)\delta^{-d}$.

[Hint: To prove (ii), use Lemma 1.2 in Chapter 3 to find a collection of disjoint balls $B_1, B_2, \ldots, B_N$ of radius $\delta/3$, each centered at E, such that their "triples" $\tilde{B}_1, \tilde{B}_2, \ldots, \tilde{B}_N$ (of radius δ) cover E. Then $\#(\delta) \leq N$, while $Nm(B_j) = cN\delta^d \leq m(E^\delta)$, since the balls B_j are disjoint and are contained in E^δ.]

18. Let E be a compact set in $\mathbb{R}^d$.

(a) Prove that $\dim(E) \leq \dim_M(E)$, where dim and $\dim_M$ are the Hausdorff and Minkowski dimensions, respectively.

(b) However, prove that if $E = \{0, 1/\log 2, 1/\log 3, \ldots, 1/\log n, \ldots\}$, then $\dim_M E = 1$, yet $\dim E = 0$.

19. Show that there is a constant c_d, dependent only on the dimension d, such that whenever E is a compact set,

$$m(E^{2\delta}) \leq c_d m(E^{\delta}).$$

[Hint: Consider the maximal function f^*, with $f = \chi_{E^{\delta}}$, and take $c_d = 6^d$.]

20. Show that if F is the self-similar set considered in Theorem 2.12, then it has the same Minkowski dimension as Hausdorff dimension.

[Hint: Each F_k is the union of m^k balls of radius cr^k. In the converse direction one sees by Lemma 2.13 that if $\epsilon = r^k$, then each ball of radius ϵ can contain at most c' vertices of the kth generation. So it takes at least m^k/c' such balls to cover F.]

21. From the unit interval, remove the second and fourth quarters (open intervals). Repeat this process in the remaining two closed intervals, and so on. Let F be the limiting set, so that

$$F = \{x : \ x = \sum_{k=1}^{\infty} a_k/4^k \quad a_k = 0 \text{ or } 2\}.$$

Prove that $0 < m_{1/2}(F) < \infty$.

22. Suppose F is the self-similar set arising in Theorem 2.9.

(a) Show that if $m \leq 1/r^d$, then $m_d(F_i \cap F_j) = 0$ if $i \neq j$.

(b) However, if $m \geq 1/r^d$, prove that $F_i \cap F_j$ is not empty for some $i \neq j$.

(c) Prove that under the hypothesis of Theorem 2.12

$$m_\alpha(F_i \cap F_j) = 0, \qquad \text{with } \alpha = \log m / \log(1/r), \text{ whenever } i \neq j.$$

23. Suppose $S_1, \ldots, S_m$ are similarities with ratio r, $0 < r < 1$. For each set E, let

$$\tilde{S}(E) = S_1(E) \cup \cdots \cup S_m(E),$$

and suppose F denotes the unique non-empty compact set with $\tilde{S}(F) = F$.

(a) If $\overline{x} \in F$, show that the set of points $\{\tilde{S}^n(\overline{x})\}_{n=1}^{\infty}$ is dense in F.

(b) Show that F is **homogeneous** in the following sense: if $x_0 \in F$ and B is any open ball centered at x_0, then $F \cap B$ contains a set similar to F.

24. Suppose E is a Borel subset of $\mathbb{R}^d$ with $\dim E < 1$. Prove that E is totally disconnected, that is, any two distinct points in E belong to different connected components.

[Hint: Fix $x, y \in E$, and show that $f(t) = |t - x|$ is Lipschitz of order 1, and hence $\dim f(E) < 1$. Conclude that $f(E)$ has a dense complement in $\mathbb{R}$. Pick r in the complement of $f(E)$ so that $0 < r < f(y)$, and use the fact that $E = \{t \in E : |t - x| < r\} \cup \{t \in E : |t - x| > r\}$.]

25. Let $F(t)$ be an arbitrary non-negative measurable function on $\mathbb{R}$, and $\gamma \in S^{d-1}$. Then there exists a measurable set E in $\mathbb{R}^d$, such that $F(t) = m_{d-1}(E \cap \mathcal{P}_{t,\gamma})$.

26. Theorem 4.1 can be refined for $d \geq 4$ as follows.

Define $C^{k,\alpha}$ to be the class of functions $F(t)$ on $\mathbb{R}$ that are C^k and for which $F^{(k)}(t)$ satisfies a Lipschitz condition of exponent α.

If E has finite measure, then for a.e. $\gamma \in S^{d-1}$ the function $m(E \cap \mathcal{P}_{t,\gamma})$ is in $C^{k,\alpha}$ for $k = (d-3)/2$, $\alpha < 1/2$, if d is odd, $d \geq 3$; and for, $k = (d-4)/2$, $\alpha < 1$, if d is even, $d \geq 4$.

27. Show that the modification of the inequality (2) of Theorem 4.5 fails if we drop $\|f\|_{L^2(\mathbb{R}^d)}$ from the right-hand side.

[Hint: Consider $\mathcal{R}^*(f_\epsilon)$, with f_ϵ defined by $f_\epsilon(x) = (|x| + \epsilon)^{-d+\delta}$, for $|x| \leq 1$, with δ fixed, $0 < \delta < 1$, and $\epsilon \to 0$.]

28. Construct a compact set $E \subset \mathbb{R}^d$, $d \geq 3$, such that $m_d(E) = 0$, yet E contains translates of any segment of unit length in $\mathbb{R}^d$. (While particular examples of such sets can be easily obtained from the case $d = 2$, the determination of the least Hausdorff dimension among all such sets is an open problem.)